

UMC 0.11um RFCMOS AI Advanced Enhancement High Gain 3.3V SPICE Model Document

Version 0.2 phase1

2012/05/29

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1 Introduction

1.1 Revision history

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2012/05/29	Daniel Wan	1. Original release 2. Silicon-based model

1.2 General Information

Following Logic/MM model is used as the main body of the subckt RF MOSFET model and it would be fine-tuned in order to improve Gm and Rout characteristics.

l110ae_hg33_v021

More information about the model is specified below.

Model File Name:

Spectre: l110ae_hg33_rf_v021.lib.scs, l110ae_hg33_rf_v021.mdl.scs

Hspice: l110ae_hg33_rf_v021.lib, l110ae_hg33_rf_v021.mdl

Eldo: l110ae_hg33_rf_v021.lib.eldo, l110ae_hg33_rf_v021.mdl.eldo

Inline Sub-circuit Model Name & Node Definition:

Twin Well NMOS: n_33_hgrf (drain gate source body)

PMOS: p_33_hgrf (drain gate source body psub)

Triple Well NMOS: n_33bpw_hgrf (drain gate source body nwell psub)

Corner Section:

tt : Typical n-ch MOSFET and Typical p-ch MOSFET

ff : Fast n-ch MOSFET and Fast p-ch MOSFET

ss : Slow n-ch MOSFET and Slow p-ch MOSFET

fnsp : Fast n-ch MOSFET and Slow p-ch MOSFET

snfp : Slow n-ch MOSFET and Fast p-ch MOSFET

Input Parameters:

lf : gate finger length in m

wf : gate finger width in m

nf : gate finger number

Geometry scaling:

NMOS:

Finger Length (lf): 0.34um ~ 1um

Finger Width (wf): 3um ~ 12um

Finger No (nf): 2~ 32

Support even and odd finger number

PMOS:

Finger Length (lf): 0.34um ~ 1um

Finger Width (wf): 2.5um ~ 15um

Finger No (nf): 2 ~ 32

Support even and odd finger number

Triple-well NMOS:

Finger Length (lf): 0.34um ~ 1um

Finger Width (wf): 3um ~ 12um

Finger No (nf): 2~ 32

Support even and odd finger number

Biasing coverage:

Voltage: Upto Vdd=3.3V

Frequency: upto 20.1GHz

Temperature coverage:

-50 C ~ +125 C

Simulator Versions:

Synopsys HSPICE release 2007.09

Cadence SPECTRE version 10.1.1.111.isr8 32 bit

Mentor Graphics ELDO version 6.11_1.1

Wafer Information:

Table 1-2 Wafer Information

Mask ID	BSW17A
Lot ID	D5RPR
Wafer Number	18

1.3 Model Usage:

1.3.1 For Cadence Spectre

- Copy ModelFileName.lib.scs and ModelFileName.mdl.scs into your simulation directory.
- Add the following statement in the input netlist to the simulator
include "./ModelFileName.lib.scs " section = CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
- SetOption1 options scale=0.9
- Follow the example below to call the model
x1 (drain gate source body) n_33_hgrf lf=0.34e-6 wf=3.6e-6 nf=4 m=8

1.3.2 For Synopsis Hspice

- Copy ModelFileName.lib and ModelFileName.mdl into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Follow the example below to call the model
x1 drain gate source body n_33_hgrf lf=0.34e-6 wf=3.6e-6 nf=4 m=8

1.3.3 For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo and ModelFileName.mdl.eldo & into your simulation directory.
- Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib " CornerSection, where CornerSection could be tt, ss, ff, snfp, fnsp.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Follow the example below to call the model
x1 drain gate source body n_33_hgrf lf=0.34e-6 wf=3.6e-6 nf=4 m=8

ModelFileName: l110ae_hg33_rf_v021

1.4 Device Architecture and Layout

1.4.1 Twin Well NMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.14um.

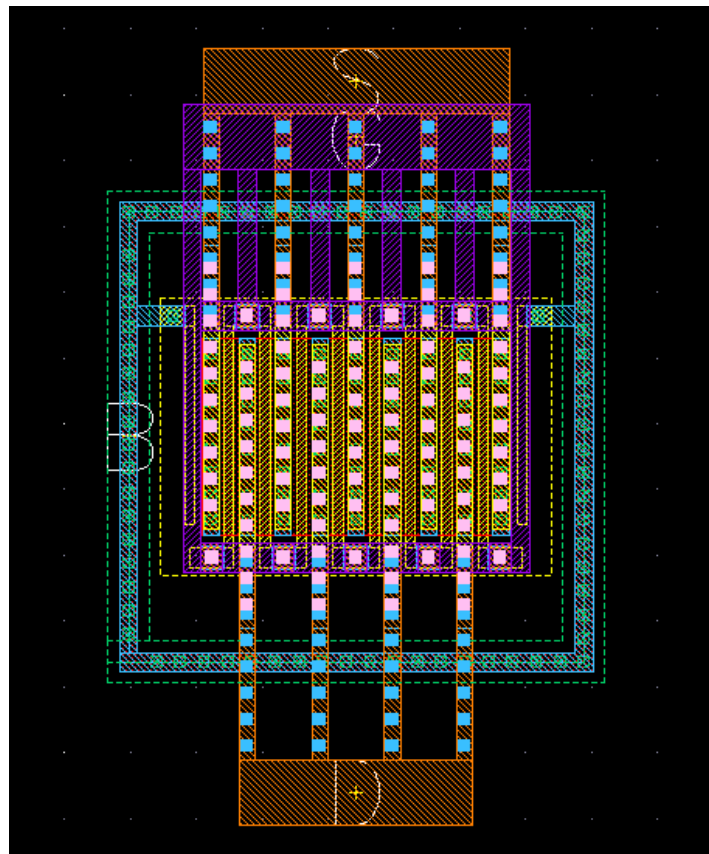


Figure 1-1 Example layout of 3.3V pwell RF NMOS

Extension equivalent circuit model:

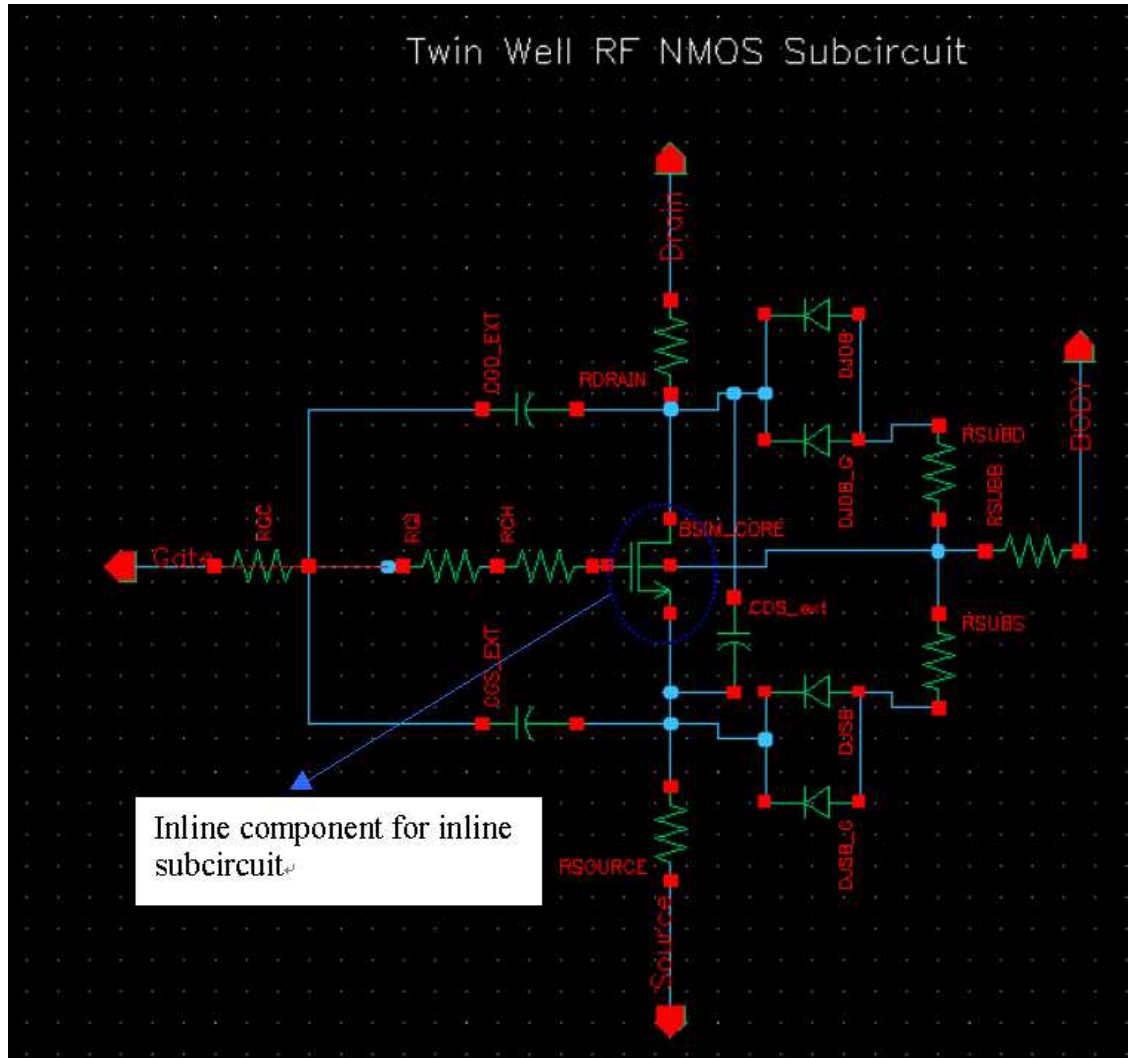


Figure 1-2 Extension equivalent circuit of TWIN-Well NMOS

1.4.2 PMOS

- Double-side gate contacts are used to reduce gate resistance.
- Dummy Poly gates are used at each side to improve the Poly CD uniformity of poly fingers.
- The pick-ups of source and drain use M1-M3 metal stacks to improve current handling capabilities.
- The space between Poly gate and S/D contacts is 0.14um.

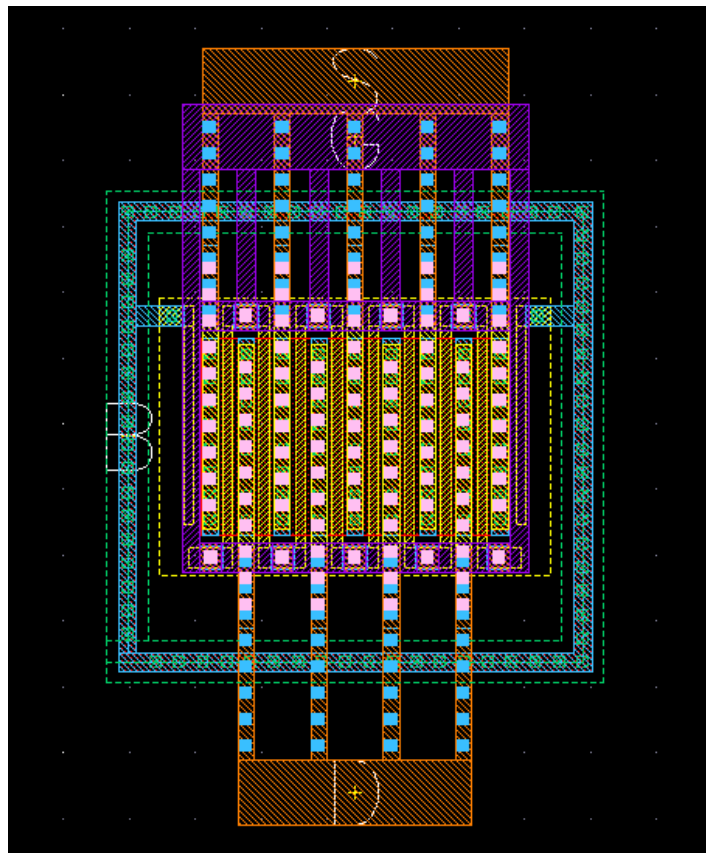


Figure 1-3 Example layout of 3.3V pwell RF NMOS

Extension equivalent circuit model:

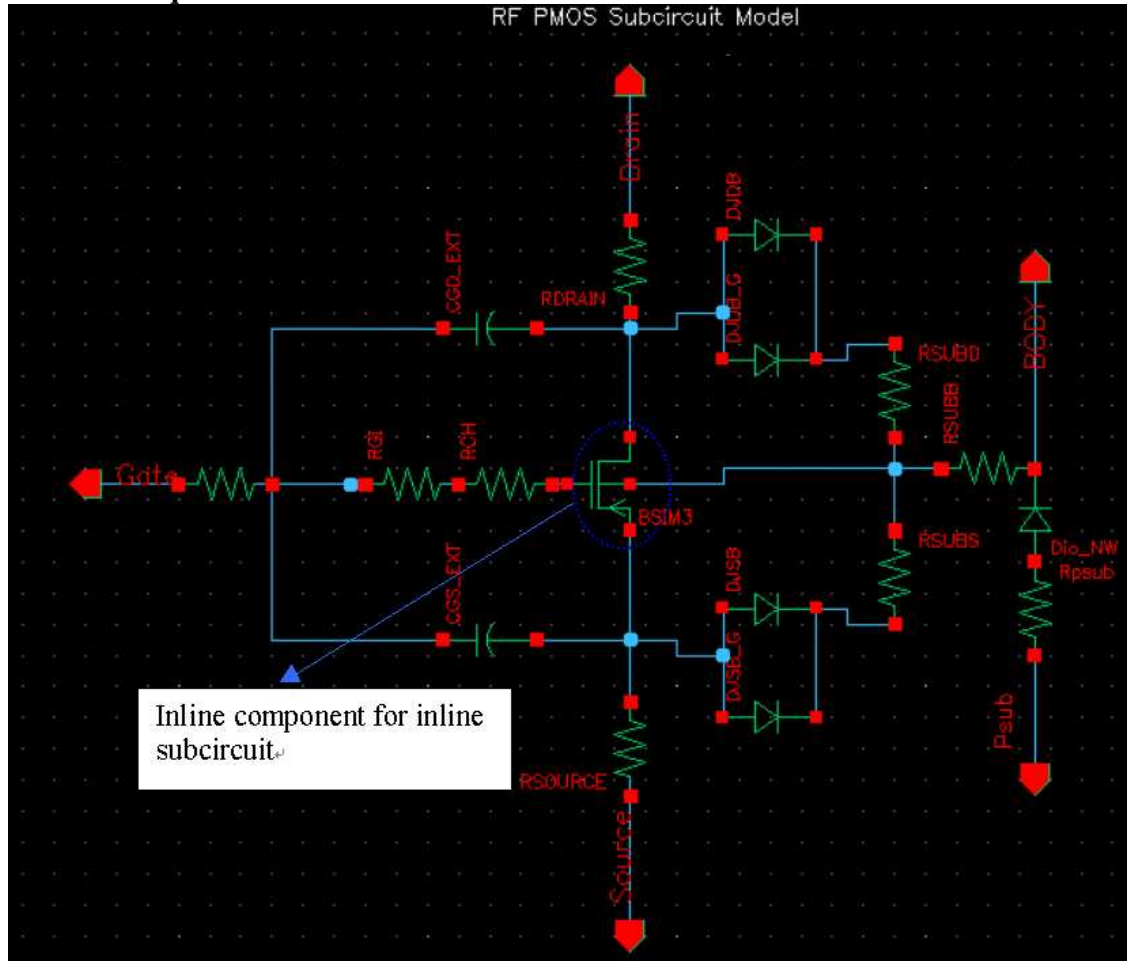


Figure 1-4 Extension equivalent circuit of PMOS

1.5 Model Features

BSIM3 :

The BSIM3 model in RF subcircuit model is based on Logic/MM model, with parameters refined to improve the fitting quality of G_m / R_{out} and RF performance.

LOD effect :

LOD model has been added to model STI stress & Finger number effects.

1/f Noise :

The BSIM3 1/f noise model is included

Note:

Model can support odd finger numbers, but it is recommended to use even finger number to reduce drain junction capacitance.

It is recommended not to use the model outside the above WF, LF and NF range. Outside above range, the model fitting quality may become worse.

This model should be used in conjunction with RF FET Pcell (Programmable layout Cell) provided in UMC MM/RF FDK (Foundry Design Kit). The model may not be valid if the transistor layout is different from UMC Pcell as shown in above layout example.

2 RF Model Verification

2.1 Twin Well NMOS Model

2.1.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.

The biasing conditions for the parameters are:

ft : Cut-off frequency extracted @ Ft_max : Vd=3.3V, sweep Vg
vth : Linear Vth extracted @ Vd=0.05V @
Id=300nA*WF*0.9*NF/(LF*0.9+0.012um)
vtsat : Saturation Vth extracted @ Vd=3.3V using fixed current
Id=300nA*WF*0.9*NF/(LF*0.9+0.012um)
body effect : vtsat shift for Vbody =0 ~ -3.3V
Idsat : Drain current extracted @ Vd=Vg=3.3V
Idlin : Drain current extracted @ Vd=0.05V, Vg=3.3V
Ioff : Drain current extracted @ Vg=0V, Vd=3.3V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	12	12	6	3	12	12	6	3
	NF	16	16	16	16	16	16	16	16
	M	1	1	1	1	1	1	1	1
SS	ft(GHz)	5.408	16.149	15.934	15.217	NA	NA	NA	NA
	idsat(uA/um)	2.55E+02	4.02E+02	4.05E+02	4.13E+02	2.81E+02	4.61E+02	4.59E+02	4.57E+02
	vtlin(V)	6.38E-01	7.01E-01	7.01E-01	7.01E-01	6.34E-01	6.89E-01	6.89E-01	6.90E-01
	vtsat(V)	5.90E-01	6.36E-01	6.36E-01	6.36E-01	5.92E-01	6.24E-01	6.24E-01	6.24E-01
	body_effect(V)	7.51E-01	6.10E-01	6.08E-01	6.05E-01	7.48E-01	6.03E-01	6.02E-01	6.00E-01
	ioff(pA/um)	4.38E-02	3.42E-02	3.52E-02	3.87E-02	1.02E-01	1.08E-01	1.68E-01	2.90E-01
SNFP	ft(GHz)	5.7793	18.172	17.946	17.166	NA	NA	NA	NA
	idsat(uA/um)	2.78E+02	4.48E+02	4.52E+02	4.62E+02	3.07E+02	5.19E+02	5.17E+02	5.17E+02
	vtlin(V)	6.09E-01	6.56E-01	6.55E-01	6.53E-01	6.06E-01	6.43E-01	6.43E-01	6.42E-01
	vtsat(V)	5.63E-01	5.91E-01	5.91E-01	5.89E-01	5.64E-01	5.78E-01	5.77E-01	5.76E-01
	body_effect(V)	7.28E-01	5.73E-01	5.72E-01	5.69E-01	7.26E-01	5.66E-01	5.65E-01	5.64E-01
	ioff(pA/um)	8.31E-02	1.04E-01	1.07E-01	1.13E-01	1.40E-01	2.12E-01	2.73E-01	3.99E-01
TT	ft(GHz)	6.3323	20.442	20.145	19.225	NA	NA	NA	NA
	idsat(uA/um)	3.05E+02	5.03E+02	5.06E+02	5.17E+02	3.33E+02	5.78E+02	5.76E+02	5.75E+02
	vtlin(V)	5.82E-01	6.11E-01	6.09E-01	6.06E-01	5.78E-01	5.98E-01	5.97E-01	5.95E-01
	vtsat(V)	5.36E-01	5.46E-01	5.45E-01	5.43E-01	5.38E-01	5.32E-01	5.32E-01	5.29E-01
	body_effect(V)	7.27E-01	5.65E-01	5.63E-01	5.61E-01	7.25E-01	5.58E-01	5.57E-01	5.55E-01
	ioff(pA/um)	1.78E-01	3.72E-01	3.82E-01	4.12E-01	2.33E-01	6.13E-01	6.85E-01	8.41E-01
FNFP	ft(GHz)	6.8757	22.765	22.407	21.355	NA	NA	NA	NA
	idsat(uA/um)	3.31E+02	5.57E+02	5.61E+02	5.73E+02	3.59E+02	6.37E+02	6.34E+02	6.34E+02
	vtlin(V)	5.54E-01	5.65E-01	5.64E-01	5.60E-01	5.51E-01	5.53E-01	5.51E-01	5.48E-01
	vtsat(V)	5.10E-01	5.01E-01	5.00E-01	4.96E-01	5.11E-01	4.86E-01	4.85E-01	4.82E-01
	body_effect(V)	7.26E-01	5.56E-01	5.54E-01	5.52E-01	7.24E-01	5.49E-01	5.48E-01	5.47E-01
	ioff(pA/um)	3.81E-01	1.36E+00	1.41E+00	1.56E+00	4.31E-01	2.11E+00	2.24E+00	2.57E+00
FF	ft(GHz)	7.1566	25.21	24.803	23.628	NA	NA	NA	NA
	idsat(uA/um)	3.53E+02	6.07E+02	6.11E+02	6.26E+02	3.84E+02	6.95E+02	6.93E+02	6.94E+02
	vtlin(V)	5.27E-01	5.20E-01	5.17E-01	5.12E-01	5.24E-01	5.07E-01	5.05E-01	5.01E-01
	vtsat(V)	4.84E-01	4.54E-01	4.52E-01	4.48E-01	4.86E-01	4.39E-01	4.37E-01	4.32E-01
	body_effect(V)	7.04E-01	5.17E-01	5.16E-01	5.14E-01	7.02E-01	5.10E-01	5.09E-01	5.09E-01
	ioff(pA/um)	7.12E-01	4.58E+00	4.84E+00	5.54E+00	7.55E-01	7.23E+00	7.67E+00	8.82E+00

Table 2-1 RF & Logic Corner Table at 25 °C

Temp=125C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	12	12	6	3	12	12	6	3
	NF	16	16	16	16	16	16	16	16
	M	1	1	1	1	1	1	1	1
SS	ft(GHz)	4.414	13.919	13.765	13.223	NA	NA	NA	NA
	idsat(uA/um)	1.94E+02	3.34E+02	3.37E+02	3.45E+02	2.15E+02	3.88E+02	3.86E+02	3.84E+02
	vtlin(V)	5.69E-01	6.45E-01	6.45E-01	6.44E-01	5.63E-01	6.30E-01	6.31E-01	6.31E-01
	vtsat(V)	5.01E-01	5.54E-01	5.54E-01	5.54E-01	5.01E-01	5.41E-01	5.41E-01	5.41E-01
	body_effect(V)	7.70E-01	5.98E-01	5.96E-01	5.93E-01	7.67E-01	5.90E-01	5.89E-01	5.87E-01
	ioff(pA/um)	1.56E+01	1.48E+01	1.47E+01	1.47E+01	1.54E+01	1.96E+01	1.95E+01	1.95E+01
SNFP	ft(GHz)	4.7454	15.754	15.598	15.016	NA	NA	NA	NA
	idsat(uA/um)	2.13E+02	3.75E+02	3.78E+02	3.88E+02	2.36E+02	4.40E+02	4.38E+02	4.37E+02
	vtlin(V)	5.38E-01	5.96E-01	5.96E-01	5.94E-01	5.32E-01	5.82E-01	5.82E-01	5.81E-01
	vtsat(V)	4.72E-01	5.08E-01	5.08E-01	5.06E-01	4.73E-01	4.94E-01	4.94E-01	4.92E-01
	body_effect(V)	7.48E-01	5.57E-01	5.56E-01	5.53E-01	7.45E-01	5.50E-01	5.49E-01	5.48E-01
	ioff(pA/um)	2.60E+01	3.58E+01	3.60E+01	3.71E+01	2.56E+01	4.84E+01	4.87E+01	5.00E+01
TT	ft(GHz)	5.203	17.863	17.653	16.964	NA	NA	NA	NA
	idsat(uA/um)	2.33E+02	4.23E+02	4.26E+02	4.37E+02	2.55E+02	4.91E+02	4.89E+02	4.88E+02
	vtlin(V)	5.09E-01	5.49E-01	5.48E-01	5.45E-01	5.03E-01	5.35E-01	5.34E-01	5.32E-01
	vtsat(V)	4.45E-01	4.62E-01	4.62E-01	4.59E-01	4.46E-01	4.47E-01	4.47E-01	4.44E-01
	body_effect(V)	7.47E-01	5.48E-01	5.46E-01	5.43E-01	7.44E-01	5.40E-01	5.39E-01	5.38E-01
	ioff(pA/um)	4.67E+01	9.50E+01	9.67E+01	1.02E+02	4.61E+01	1.29E+02	1.32E+02	1.38E+02
FNFP	ft(GHz)	5.6495	20.015	19.759	18.971	NA	NA	NA	NA
	idsat(uA/um)	2.53E+02	4.71E+02	4.75E+02	4.87E+02	2.75E+02	5.43E+02	5.40E+02	5.40E+02
	vtlin(V)	4.81E-01	5.02E-01	5.01E-01	4.97E-01	4.75E-01	4.88E-01	4.87E-01	4.83E-01
	vtsat(V)	4.18E-01	4.16E-01	4.15E-01	4.11E-01	4.19E-01	4.01E-01	4.00E-01	3.96E-01
	body_effect(V)	7.46E-01	5.38E-01	5.36E-01	5.34E-01	7.43E-01	5.30E-01	5.29E-01	5.28E-01
	ioff(pA/um)	8.37E+01	2.53E+02	2.60E+02	2.82E+02	8.26E+01	3.48E+02	3.58E+02	3.86E+02
FF	ft(GHz)	5.929	22.289	22.003	21.125	NA	NA	NA	NA
	idsat(uA/um)	2.72E+02	5.18E+02	5.22E+02	5.36E+02	2.96E+02	5.99E+02	5.96E+02	5.97E+02
	vtlin(V)	4.52E-01	4.55E-01	4.53E-01	4.47E-01	4.46E-01	4.41E-01	4.39E-01	4.34E-01
	vtsat(V)	3.92E-01	3.69E-01	3.67E-01	3.62E-01	3.92E-01	3.52E-01	3.51E-01	3.46E-01
	body_effect(V)	7.23E-01	4.94E-01	4.93E-01	4.91E-01	7.20E-01	4.87E-01	4.86E-01	4.86E-01
	ioff(pA/um)	1.37E+02	6.42E+02	6.69E+02	7.43E+02	1.35E+02	9.08E+02	9.43E+02	1.05E+03

Table 2-2 RF & Logic Corner Table at 125 °C

Temp=-50C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	12	12	6	3	12	12	6	3
	NF	16	16	16	16	16	16	16	16
	M	1	1	1	1	1	1	1	1
SS	ft(GHz)	6.197	17.616	17.336	16.447	NA	NA	NA	NA
	idsat(uA/um)	3.12E+02	4.50E+02	4.52E+02	4.60E+02	3.44E+02	5.11E+02	5.09E+02	5.07E+02
	vtlin(V)	6.96E-01	7.53E-01	7.53E-01	7.53E-01	6.94E-01	7.42E-01	7.42E-01	7.43E-01
	vtsat(V)	6.62E-01	7.03E-01	7.03E-01	7.03E-01	6.64E-01	6.91E-01	6.92E-01	6.92E-01
	body_effect(V)	7.37E-01	6.20E-01	6.19E-01	6.15E-01	7.34E-01	6.13E-01	6.12E-01	6.10E-01
	ioff(pA/um)	1.97E-03	1.97E-03	2.80E-03	6.41E-03	6.05E-02	6.05E-02	1.23E-01	2.44E-01
SNFP	ft(GHz)	6.5946	19.751	19.45	18.478	NA	NA	NA	NA
	idsat(uA/um)	3.40E+02	4.98E+02	5.01E+02	5.10E+02	3.76E+02	5.72E+02	5.70E+02	5.69E+02
	vtlin(V)	6.69E-01	7.09E-01	7.09E-01	7.07E-01	6.67E-01	6.98E-01	6.97E-01	6.96E-01
	vtsat(V)	6.36E-01	6.58E-01	6.58E-01	6.56E-01	6.37E-01	6.45E-01	6.45E-01	6.44E-01
	body_effect(V)	7.15E-01	5.86E-01	5.85E-01	5.82E-01	7.12E-01	5.79E-01	5.78E-01	5.77E-01
	ioff(pA/um)	1.97E-03	1.97E-03	2.80E-03	6.25E-03	6.05E-02	6.05E-02	1.22E-01	2.44E-01
TT	ft(GHz)	7.2163	22.08	21.694	20.555	NA	NA	NA	NA
	idsat(uA/um)	3.72E+02	5.56E+02	5.58E+02	5.69E+02	4.09E+02	6.34E+02	6.32E+02	6.31E+02
	vtlin(V)	6.43E-01	6.65E-01	6.64E-01	6.61E-01	6.41E-01	6.53E-01	6.53E-01	6.51E-01
	vtsat(V)	6.10E-01	6.14E-01	6.13E-01	6.10E-01	6.11E-01	6.01E-01	6.00E-01	5.98E-01
	body_effect(V)	7.13E-01	5.79E-01	5.78E-01	5.74E-01	7.11E-01	5.72E-01	5.71E-01	5.69E-01
	ioff(pA/um)	1.97E-03	1.81E-03	2.80E-03	6.74E-03	6.05E-02	6.18E-02	1.23E-01	2.45E-01
FNFP	ft(GHz)	7.8281	24.464	24.002	22.707	NA	NA	NA	NA
	idsat(uA/um)	4.04E+02	6.13E+02	6.16E+02	6.27E+02	4.41E+02	6.96E+02	6.93E+02	6.93E+02
	vtlin(V)	6.16E-01	6.21E-01	6.19E-01	6.15E-01	6.14E-01	6.09E-01	6.08E-01	6.05E-01
	vtsat(V)	5.84E-01	5.69E-01	5.68E-01	5.64E-01	5.85E-01	5.55E-01	5.54E-01	5.51E-01
	body_effect(V)	7.13E-01	5.71E-01	5.70E-01	5.68E-01	7.11E-01	5.65E-01	5.64E-01	5.62E-01
	ioff(pA/um)	1.97E-03	2.47E-03	4.11E-03	8.06E-03	6.05E-02	6.25E-02	1.24E-01	2.47E-01
FF	ft(GHz)	8.1022	27	26.474	25.03	NA	NA	NA	NA
	idsat(uA/um)	4.29E+02	6.63E+02	6.66E+02	6.79E+02	4.69E+02	7.54E+02	7.51E+02	7.51E+02
	vtlin(V)	5.91E-01	5.76E-01	5.74E-01	5.69E-01	5.88E-01	5.64E-01	5.63E-01	5.58E-01
	vtsat(V)	5.59E-01	5.23E-01	5.21E-01	5.16E-01	5.60E-01	5.08E-01	5.06E-01	5.01E-01
	body_effect(V)	6.91E-01	5.36E-01	5.34E-01	5.33E-01	6.89E-01	5.29E-01	5.28E-01	5.27E-01
	ioff(pA/um)	2.14E-03	7.73E-03	9.54E-03	1.41E-02	6.18E-02	7.17E-02	1.34E-01	2.58E-01

Table 2-3 RF & Logic Corner Table at -50 °C

2.1.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted I_{dsat} is smaller than the real I_{dsat} for the DUT without the parasitic resistance.

Device						Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	con	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
NFET752	0.34	6	16	2	1	0.601	0.61	0.538	0.545	491.57	500.39	123.89	1.68	19.668	19.972
NFET753	0.34	12	16	1	1	0.6	0.611	0.537	0.546	468.38	492.53	121.12	1.021	19.749	20.137
NFET754	0.34	2	16	5	1	0.604	0.604	0.538	0.54	524.5	528.47	128.57	4.34	17.999	17.773
NFET755	0.34	6	4	8	1	0.606	0.61	0.542	0.545	505.15	502.96	151.41	1.909	19.607	19.157
NFET756	0.34	6	8	4	1	0.603	0.61	0.54	0.545	497.13	501.5	113.47	1.756	19.639	19.609
NFET757	0.34	6	32	1	1	0.598	0.61	0.536	0.545	487.74	499.12	111.31	1.642	19.557	20.107
NFET758	0.34	3	2	14	1	0.614	0.607	0.546	0.543	527.65	516.36	270.14	4.076	17.809	17.577
NFET759	0.34	3	4	12	1	0.61	0.607	0.545	0.543	521.87	515.61	192.6	3.465	18.473	18.414
NFET760	0.34	3	8	8	1	0.608	0.607	0.541	0.543	516.74	514.71	148.64	3.16	18.856	18.843
NFET761	0.34	3	11	6	1	0.603	0.607	0.538	0.543	515.09	513.95	138.16	3.077	18.906	18.794
NFET762	0.34	3	32	2	1	0.602	0.607	0.538	0.543	510.65	513.15	112.7	2.931	18.944	19.224
NFET763	0.42	12	16	1	1	0.603	0.609	0.557	0.553	430.59	456.45	112	0.894	16.311	16.661
NFET764	0.5	9	16	1	1	0.599	0.605	0.56	0.553	412.46	428.05	153.1	1.067	13.863	13.993
NFET765	0.7	12	16	1	1	0.592	0.594	0.557	0.547	345.03	365.09	152.5	0.827	9.595	9.726
NFET766	1	12	16	1	1	0.584	0.582	0.551	0.536	289.04	300.77	121.18	0.827	6.351	6.297
NFET768	0.34	12	4	4	1	0.608	0.611	0.544	0.546	486.02	497.5	111.2	1.136	20.059	19.431
NFET770	0.34	12	32	1	1	0.598	0.611	0.535	0.546	445.72	488.75	55.97	1.001	19.043	20.254
NFET772	0.5	12	11	2	1	0.598	0.605	0.559	0.553	398.44	426.1	78.82	0.867	13.646	13.939
NFET791	0.36	3	4	12	1	0.612	0.607	0.551	0.545	509.36	505.04	187.2	3.414	17.778	17.569

Table 2-4 Summary Table for Twin Well RF NMOS

Measure: Measurement result.

Model: Simulation result

2.1.3 Simulation results for WF, LF and NF trends

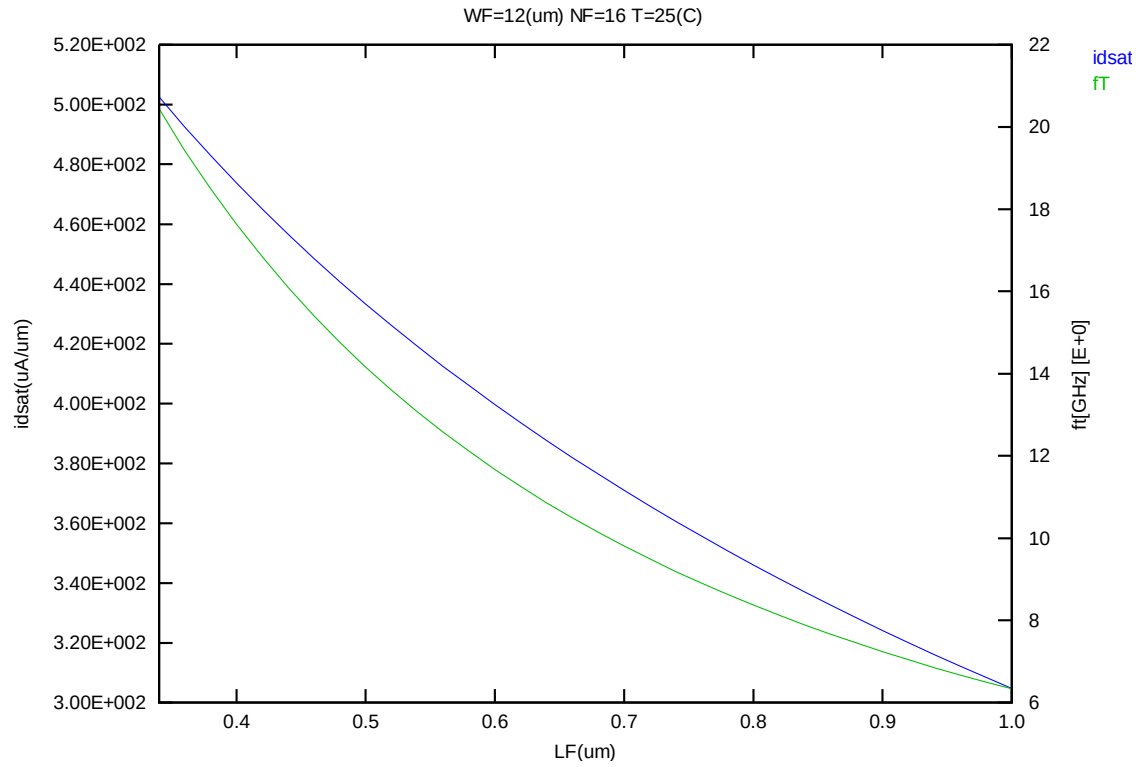
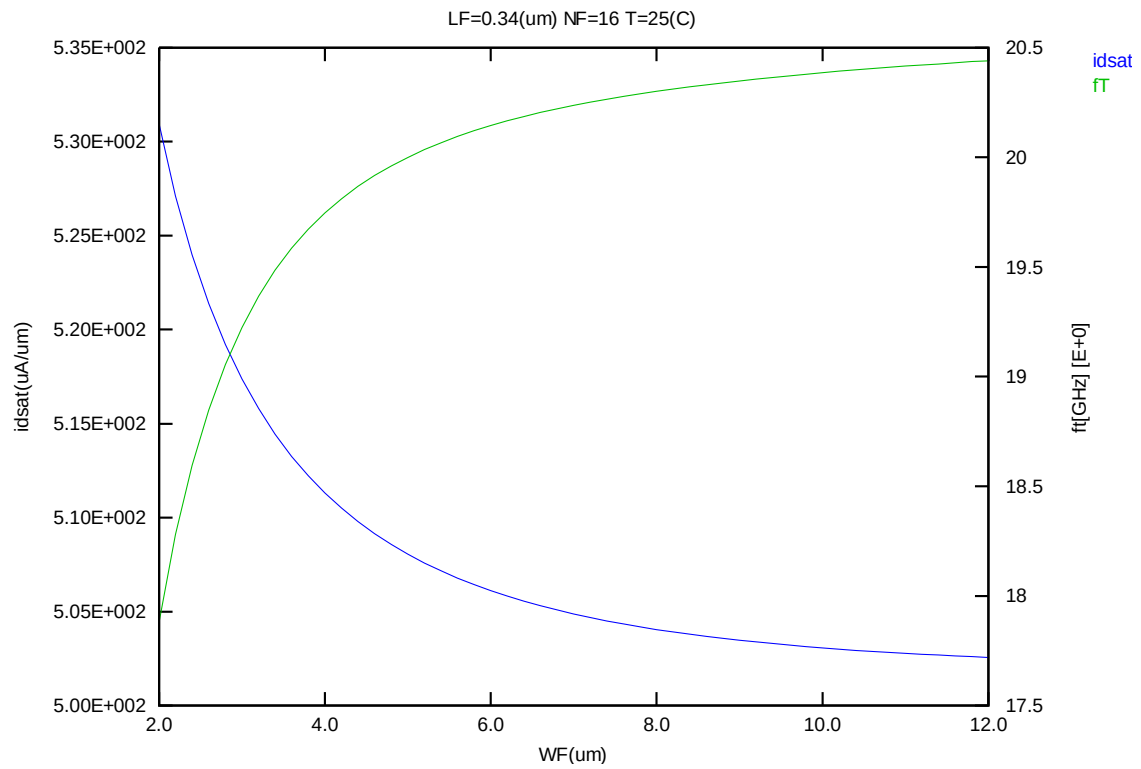


Figure 2-1 fT and Idsat vs. channel length



2.1.4 DC & RF Fitting curves for individual devices

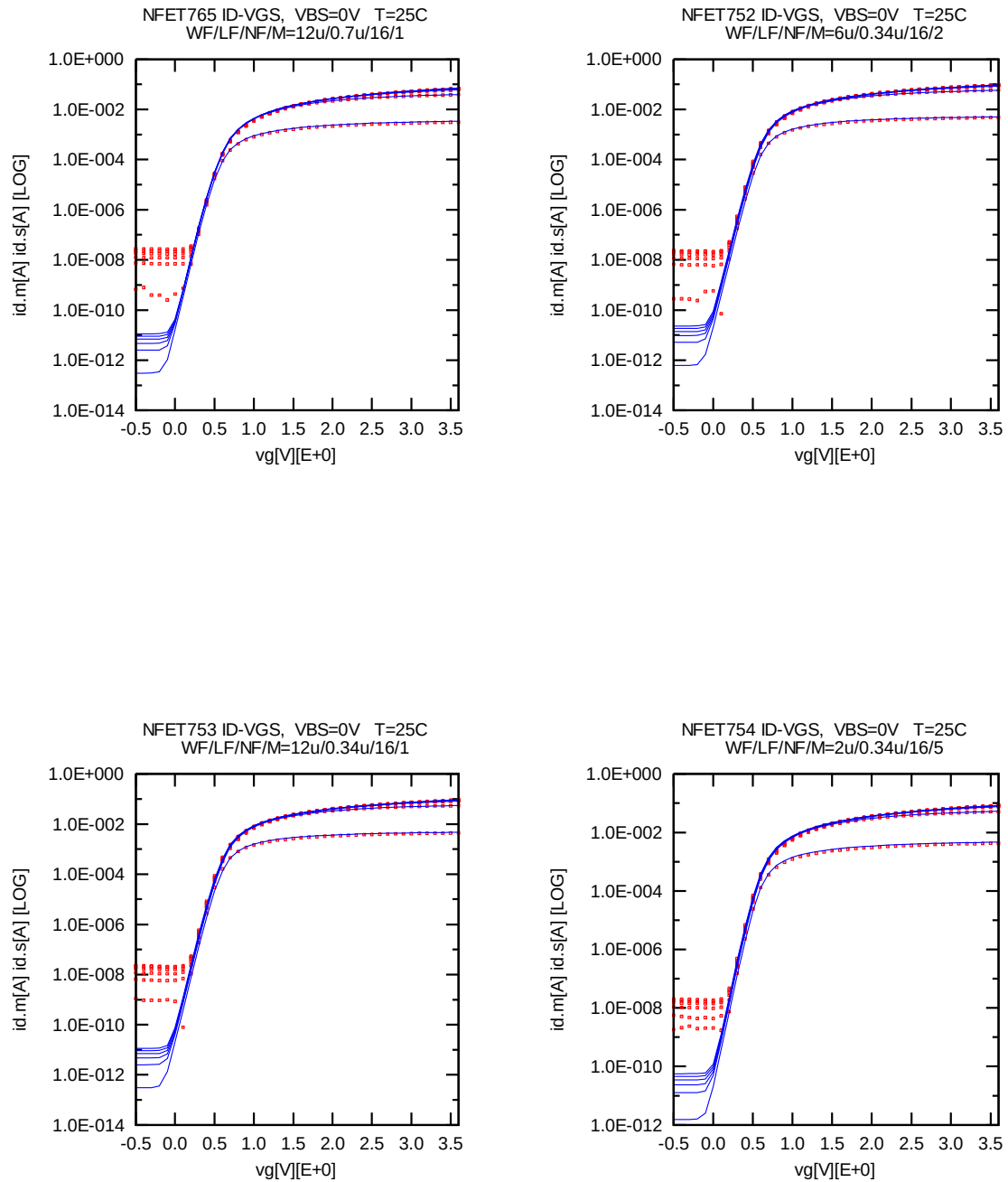
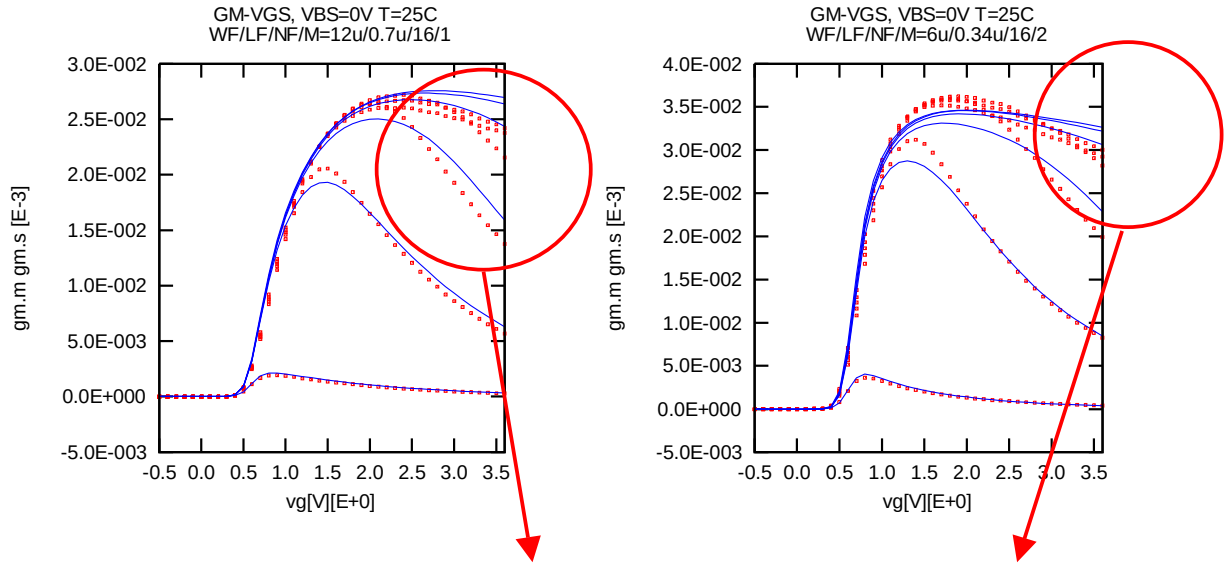


Figure 2-3 Id-Vg (25°C)



RF FET needs sufficient large total gate width in order to get smooth and reliable S-parameter measurement. However, too large a gate width will cause self-heating effect at high Vd and Vg biases, as the negative output resistance seen in the next plot. This self-heating effect will also introduce larger discrepancy between model and silicon data in transconductance.

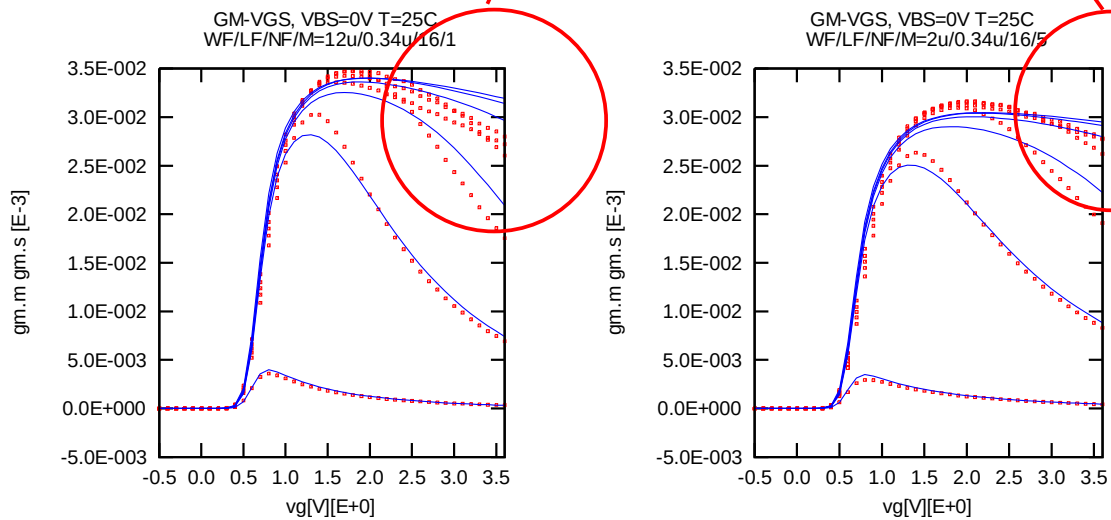
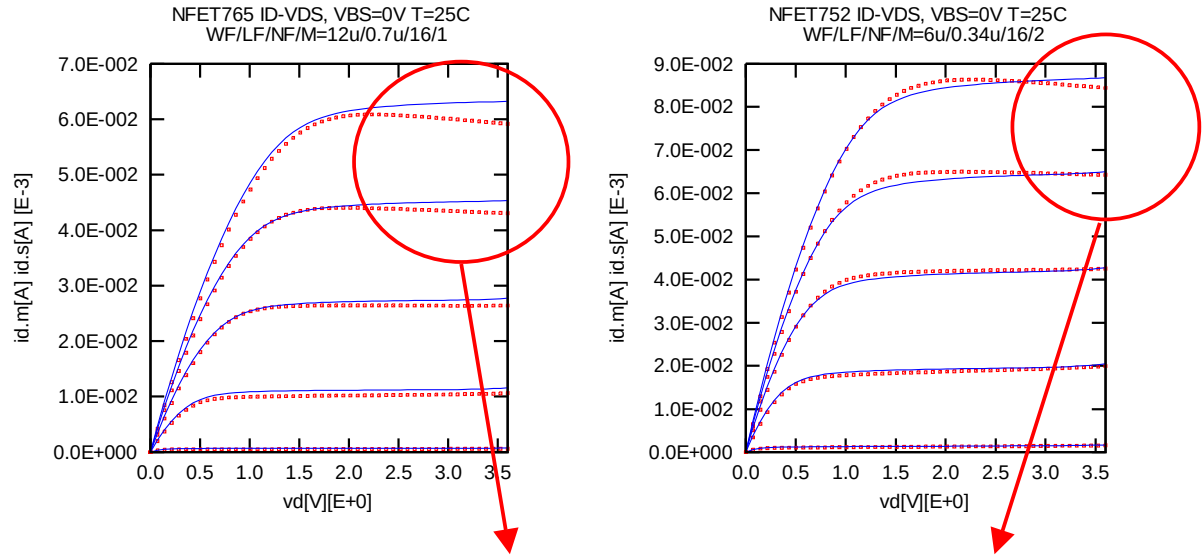


Figure 2-4 Gm-Vg (25°C)



RF FET needs sufficient large total gate width in order to get smooth and reliable S-parameter measurement. However, too large a gate width will cause self-heating effect at high Vd and Vg biases, as the negative output resistance seen in the plots.

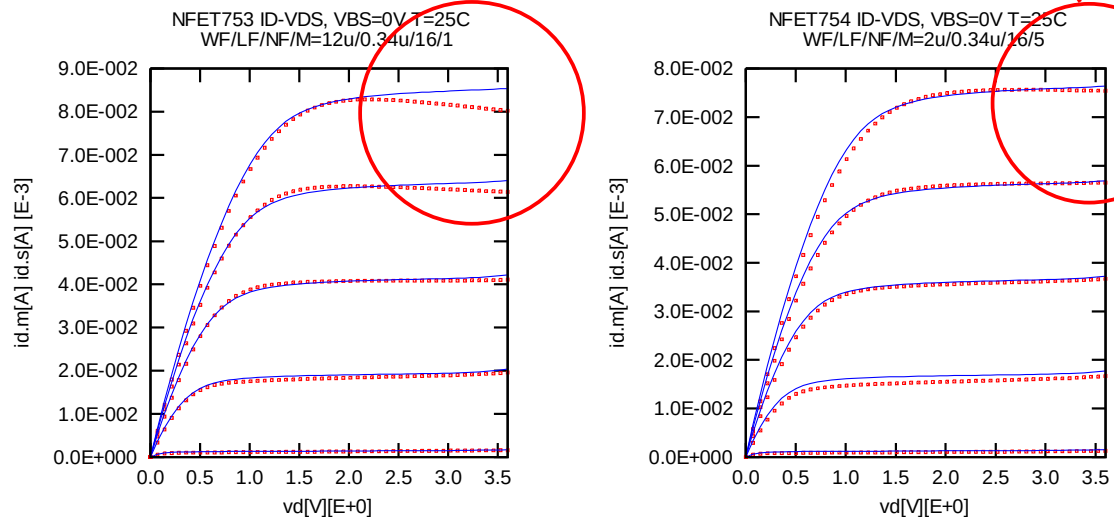
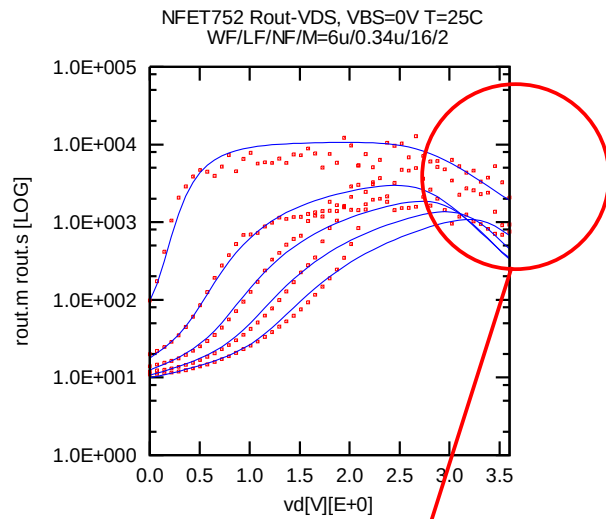
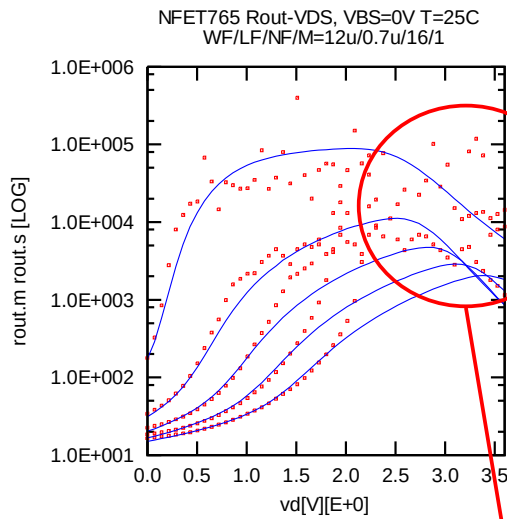


Figure 2-5 Id-Vd (25°C)



RF FET needs sufficient large total gate width in order to get smooth and reliable S-parameter measurement. However, too large a gate width will cause self-heating effect at high Vd and Vg biases, as the negative output resistance seen in the plots.

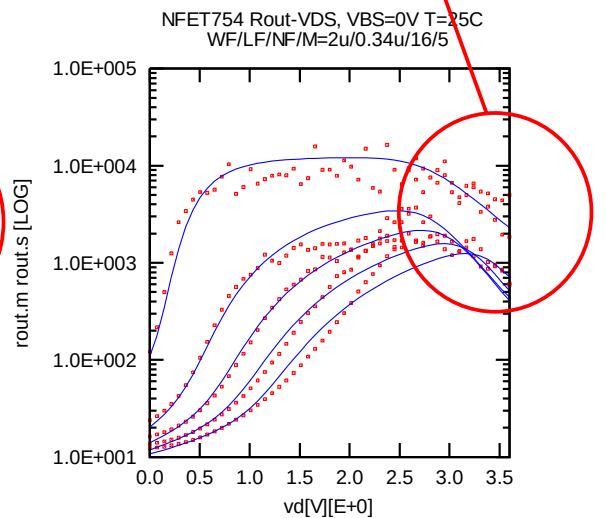
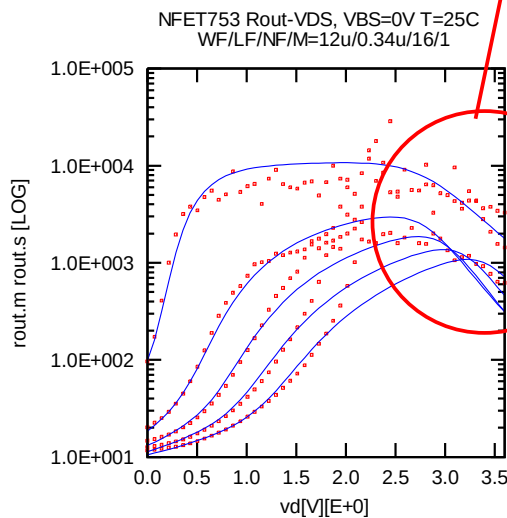


Figure 2-6 Rout-Vd (25°C)

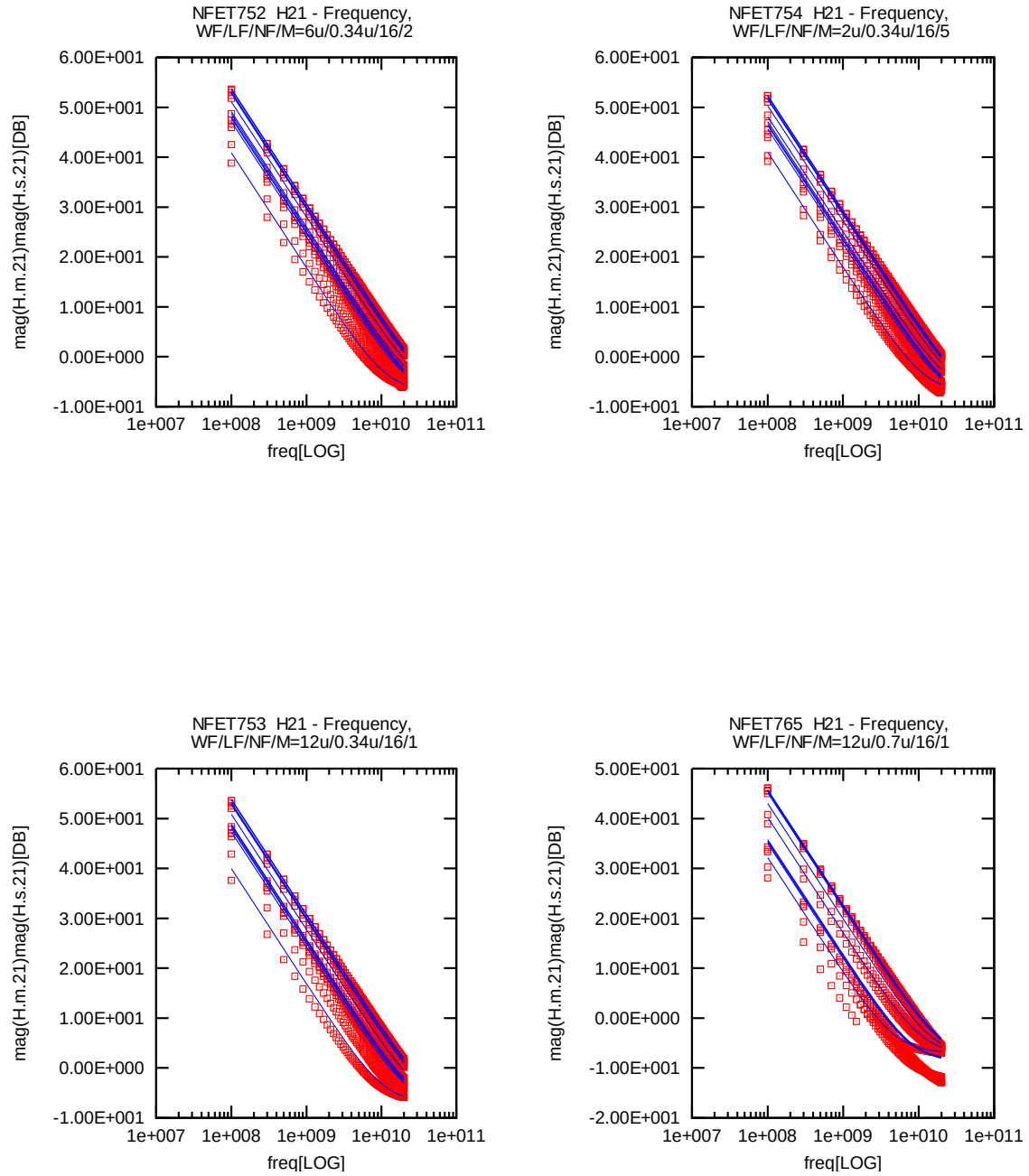


Figure 2-7 Mag(H21)-Freq (25°C)

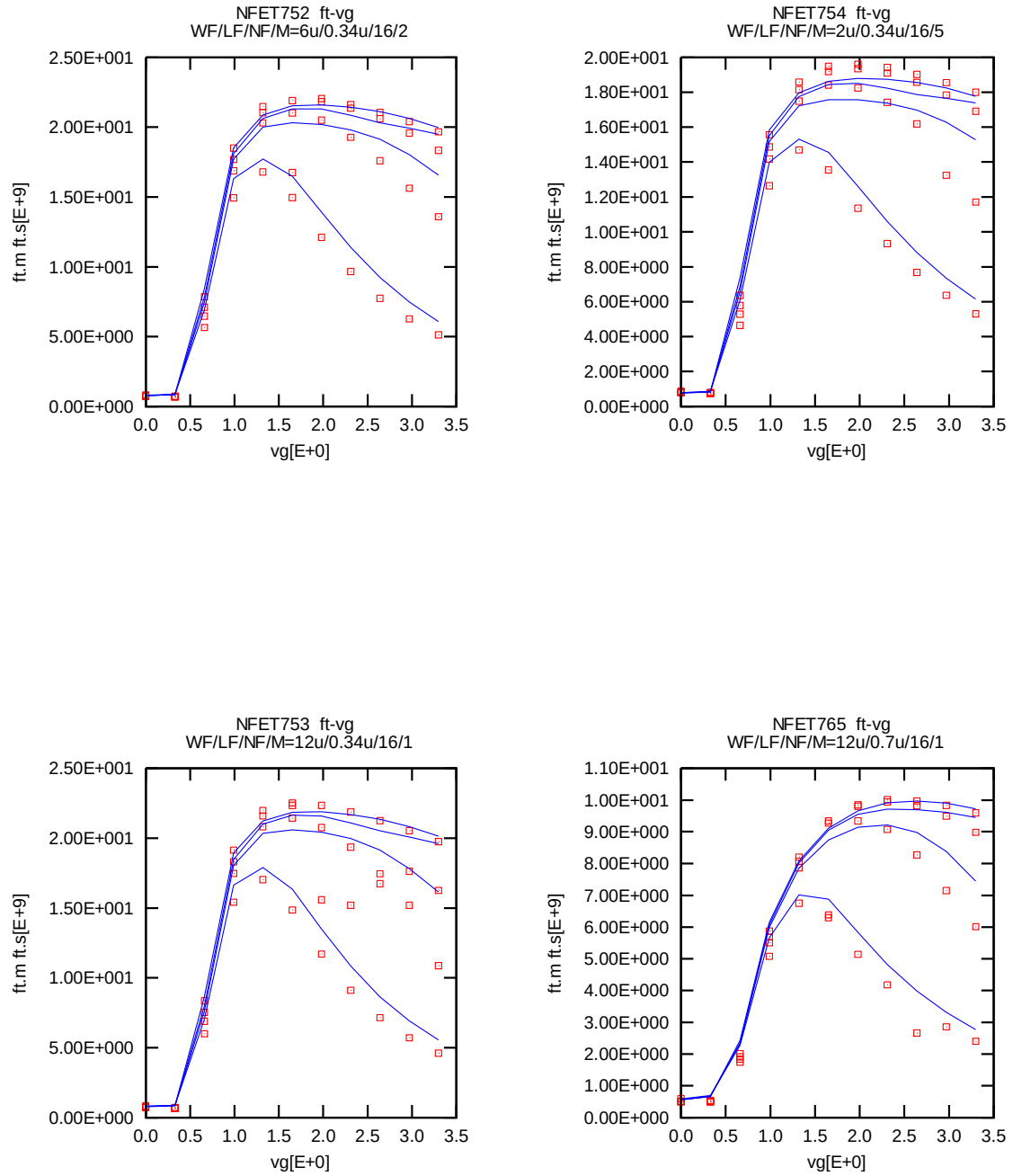


Figure 2-8 f_T - V_g (25°C)

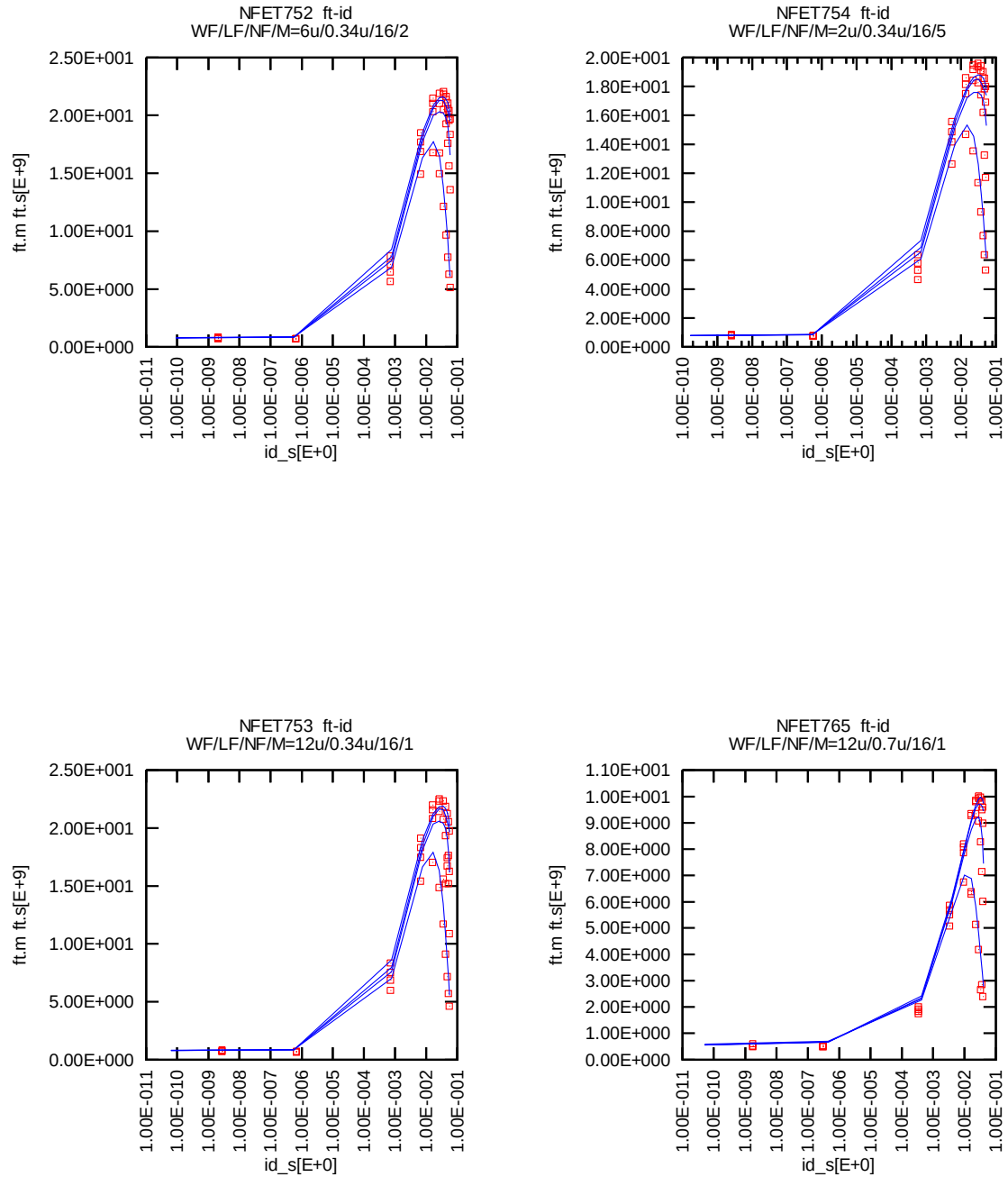
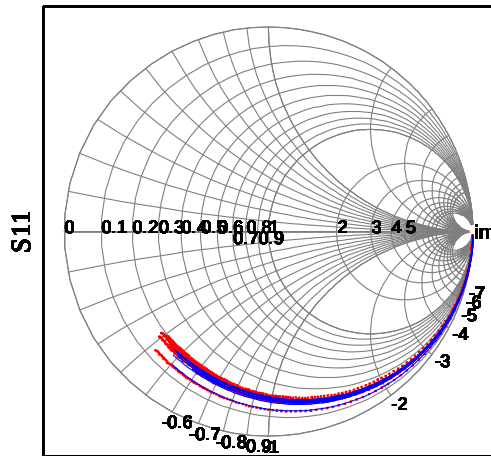
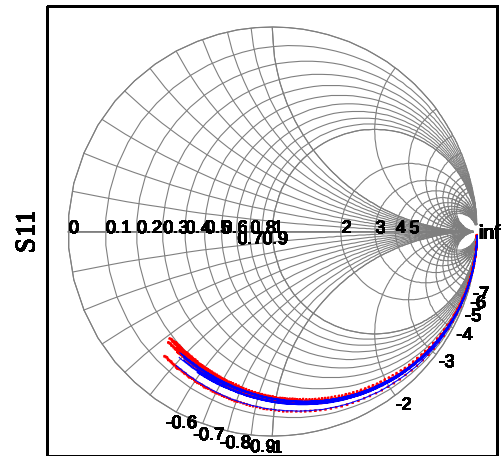


Figure 2-9 f_T -Id (25°C)

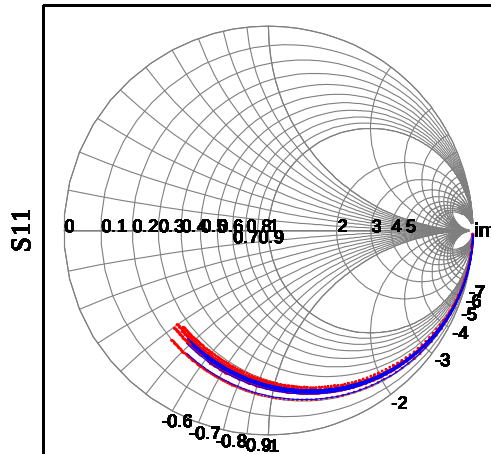
NFET752 S11 WF/LF/NF/M=6u/0.34u/16/2



NFET754 S11 WF/LF/NF/M=2u/0.34u/16/5



NFET753 S11 WF/LF/NF/M=12u/0.34u/16/1



NFET765 S11 WF/LF/NF/M=12u/0.7u/16/1

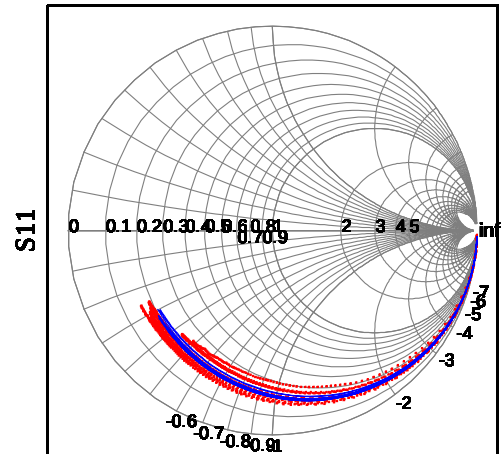


Figure 2-10 S11 (25°C)

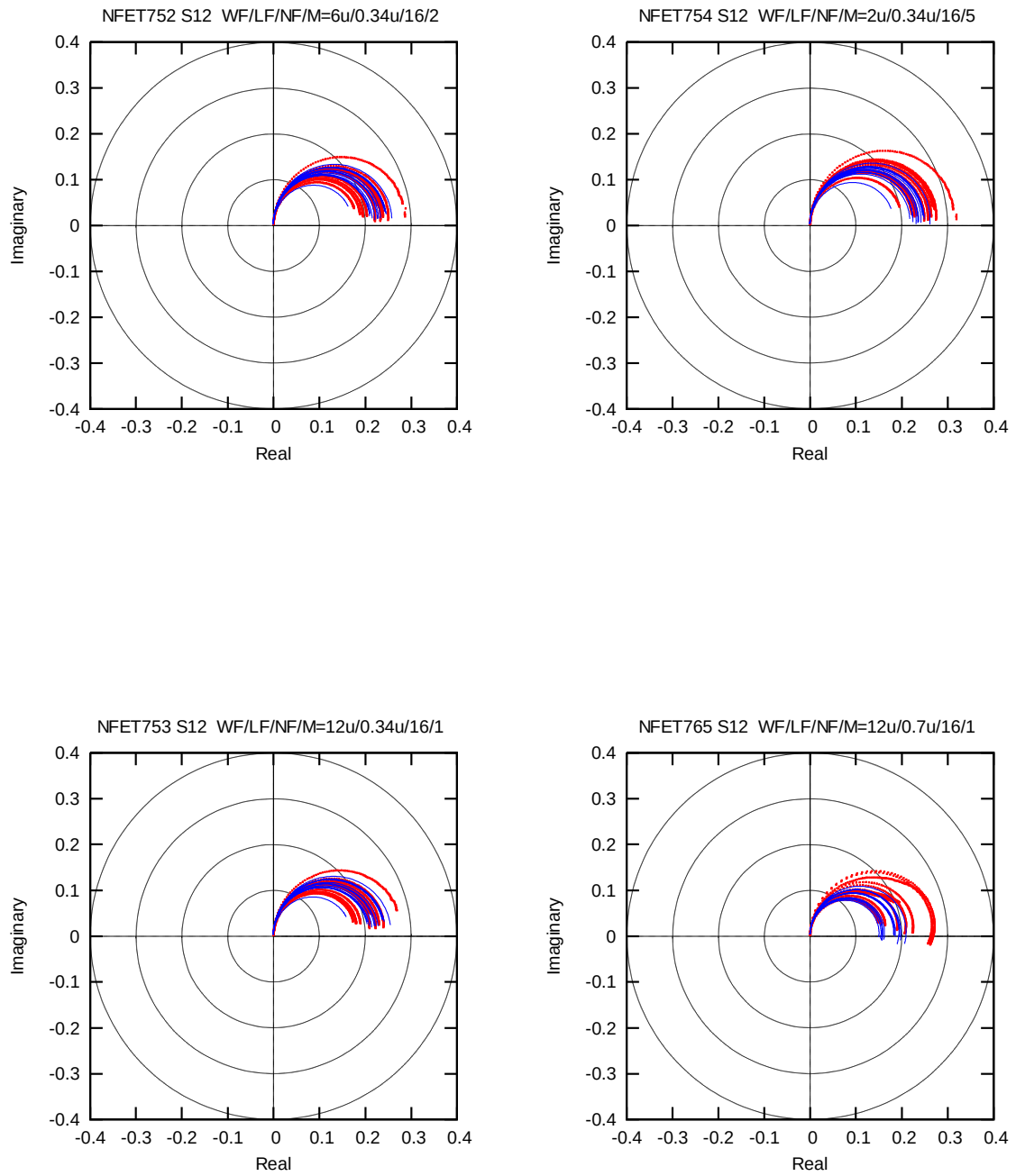


Figure 2-11 S12 (25°C)

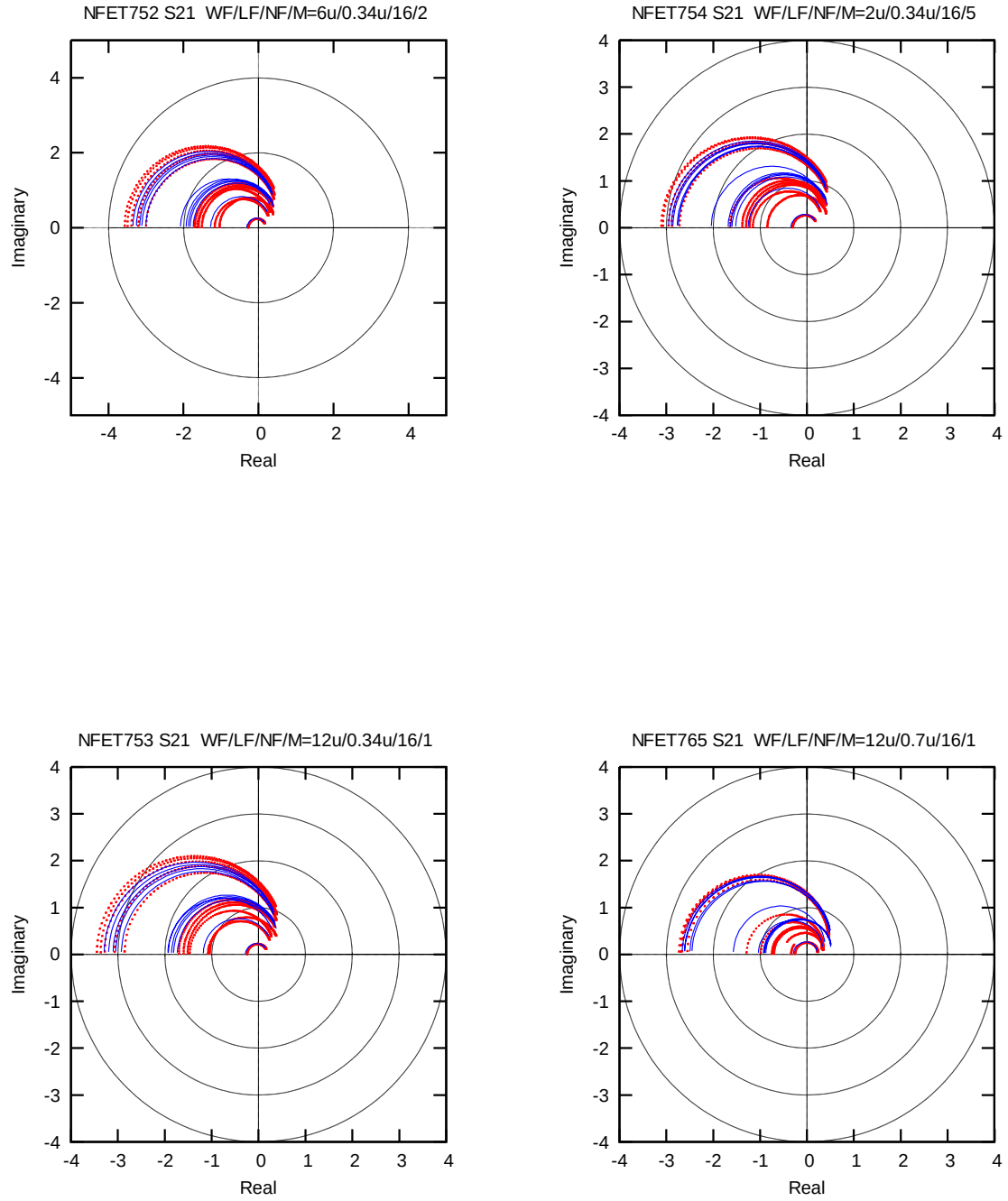


Figure 2-12 S21 (25°C)

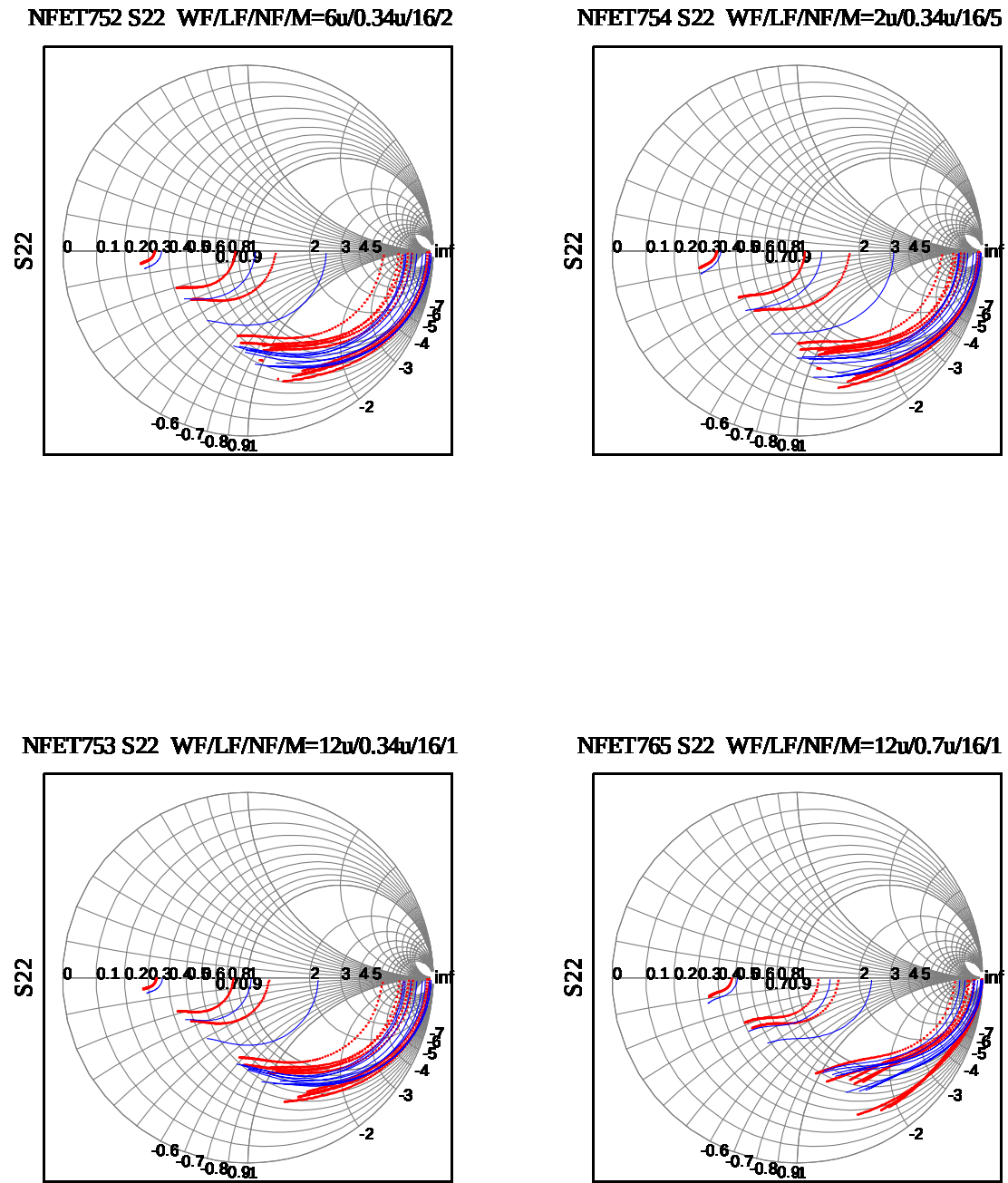


Figure 2-13 S22 (25°C)

2.2 PMOS Model

2.2.1 Corner Table

The tables below summarize the DC & RF simulation results for different corner cases for some selected devices and compared with Logic/MM model.

The biasing conditions for the parameters are:

ft : Cut-off frequency extracted @ Ft_max:Vd=-3.3V, sweep Vg
vth : Linear Vth extracted @ Vd=-0.05V @
Id=-70nA*WF*0.9*NF/(LF*0.9+0.012um)
vtsat : Saturation Vth extracted @ Vd=-3.3V using fixed current
Id=-70nA*WF*0.9*NF/(LF*0.9+0.012um)
body effect : vtsat shift for Vbody =0 ~ 3.3V
Idsat : Drain current extracted @ Vd=Vg=-3.3V
Idlin : Drain current extracted @ Vd=-0.05, Vg=-3.3V
Ioff : Drain current extracted @ Vg=0V, Vd=-3.3V

Temp=25C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	10	15	10	5	10	15	10	5
	NF	16	16	16	16	16	16	16	16
	M	1	1	1	1	1	1	1	1
SS	ft(GHz)	2.3245	10.585	10.451	10.075	NA	NA	NA	NA
	idsat(uA/um)	-8.02E+01	-2.07E+02	-2.06E+02	-2.04E+02	-7.79E+01	-2.20E+02	-2.18E+02	-2.15E+02
	vtlin(V)	-7.45E-01	-6.95E-01	-6.97E-01	-7.03E-01	-7.08E-01	-6.70E-01	-6.73E-01	-6.78E-01
	vtsat(V)	-7.18E-01	-6.52E-01	-6.54E-01	-6.60E-01	-6.81E-01	-6.27E-01	-6.29E-01	-6.34E-01
	body_effect(V)	-1.06E+00	-8.67E-01	-8.66E-01	-8.64E-01	-1.05E+00	-8.51E-01	-8.51E-01	-8.49E-01
	ioff(pA/um)	-4.34E-03	-4.74E-02	-4.58E-02	-4.18E-02	-7.89E-02	-1.37E-01	-1.57E-01	-2.19E-01
SNFP	ft(GHz)	2.6528	13.822	13.662	13.208	NA	NA	NA	NA
	idsat(uA/um)	-9.78E+01	-2.74E+02	-2.74E+02	-2.74E+02	-9.65E+01	-3.05E+02	-3.05E+02	-3.03E+02
	vtlin(V)	-6.52E-01	-5.65E-01	-5.64E-01	-5.62E-01	-6.15E-01	-5.43E-01	-5.43E-01	-5.41E-01
	vtsat(V)	-6.27E-01	-5.19E-01	-5.18E-01	-5.16E-01	-5.90E-01	-4.99E-01	-4.98E-01	-4.95E-01
	body_effect(V)	-1.03E+00	-7.60E-01	-7.59E-01	-7.59E-01	-1.02E+00	-7.44E-01	-7.44E-01	-7.44E-01
	ioff(pA/um)	-1.97E-02	-1.22E+00	-1.24E+00	-1.29E+00	-1.21E-01	-2.10E+00	-2.17E+00	-2.35E+00
TT	ft(GHz)	2.5282	12.69	12.543	12.126	NA	NA	NA	NA
	idsat(uA/um)	-9.18E+01	-2.52E+02	-2.52E+02	-2.51E+02	-9.03E+01	-2.77E+02	-2.76E+02	-2.74E+02
	vtlin(V)	-6.83E-01	-6.07E-01	-6.07E-01	-6.08E-01	-6.46E-01	-5.85E-01	-5.85E-01	-5.86E-01
	vtsat(V)	-6.58E-01	-5.62E-01	-5.62E-01	-5.62E-01	-6.21E-01	-5.40E-01	-5.40E-01	-5.40E-01
	body_effect(V)	-1.03E+00	-7.74E-01	-7.74E-01	-7.73E-01	-1.02E+00	-7.59E-01	-7.58E-01	-7.58E-01
	ioff(pA/um)	-1.03E-02	-4.20E-01	-4.17E-01	-4.07E-01	-9.59E-02	-7.74E-01	-7.95E-01	-8.59E-01
FNFP	ft(GHz)	2.3964	11.498	11.364	10.986	NA	NA	NA	NA
	idsat(uA/um)	-8.57E+01	-2.29E+02	-2.28E+02	-2.27E+02	-8.41E+01	-2.48E+02	-2.47E+02	-2.45E+02
	vtlin(V)	-7.15E-01	-6.49E-01	-6.51E-01	-6.54E-01	-6.78E-01	-6.27E-01	-6.28E-01	-6.31E-01
	vtsat(V)	-6.89E-01	-6.04E-01	-6.05E-01	-6.08E-01	-6.52E-01	-5.81E-01	-5.82E-01	-5.84E-01
	body_effect(V)	-1.03E+00	-7.88E-01	-7.87E-01	-7.86E-01	-1.02E+00	-7.72E-01	-7.72E-01	-7.71E-01
	ioff(pA/um)	-5.92E-03	-1.46E-01	-1.42E-01	-1.32E-01	-8.37E-02	-3.07E-01	-3.24E-01	-3.79E-01
FF	ft(GHz)	2.6998	14.661	14.5	14.039	NA	NA	NA	NA
	idsat(uA/um)	-1.03E+02	-2.94E+02	-2.94E+02	-2.94E+02	-1.02E+02	-3.34E+02	-3.34E+02	-3.33E+02
	vtlin(V)	-6.22E-01	-5.21E-01	-5.20E-01	-5.15E-01	-5.85E-01	-5.00E-01	-4.98E-01	-4.93E-01
	vtsat(V)	-5.98E-01	-4.73E-01	-4.71E-01	-4.66E-01	-5.61E-01	-4.52E-01	-4.50E-01	-4.44E-01
	body_effect(V)	-1.00E+00	-6.92E-01	-6.92E-01	-6.93E-01	-9.90E-01	-6.71E-01	-6.71E-01	-6.72E-01
	ioff(pA/um)	-3.36E-02	-3.57E+00	-3.74E+00	-4.16E+00	-1.62E-01	-6.13E+00	-6.48E+00	-7.36E+00

Table 2-5 RF & Logic Corner Table at 25 °C

Temp=125C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	10	15	10	5	10	15	10	5
	NF	16	16	16	16	16	16	16	16
	M	1	1	1	1	1	1	1	1
SS	ft(GHz)	2.0521	9.3949	9.2709	8.9243	NA	NA	NA	NA
	idsat(uA/um)	-6.89E+01	-1.82E+02	-1.81E+02	-1.79E+02	-6.63E+01	-1.91E+02	-1.90E+02	-1.87E+02
	vtlin(V)	-6.44E-01	-5.98E-01	-6.00E-01	-6.07E-01	-6.09E-01	-5.74E-01	-5.77E-01	-5.83E-01
	vtsat(V)	-6.06E-01	-5.42E-01	-5.44E-01	-5.50E-01	-5.71E-01	-5.19E-01	-5.21E-01	-5.26E-01
	body_effect(V)	-1.10E+00	-9.08E-01	-9.07E-01	-9.05E-01	-1.09E+00	-8.91E-01	-8.90E-01	-8.89E-01
	ioff(pA/um)	-1.36E+00	-1.66E+01	-1.59E+01	-1.41E+01	-2.86E+00	-2.61E+01	-2.51E+01	-2.25E+01
SNFP	ft(GHz)	2.3401	12.475	12.323	11.893	NA	NA	NA	NA
	idsat(uA/um)	-8.41E+01	-2.44E+02	-2.44E+02	-2.43E+02	-8.21E+01	-2.66E+02	-2.66E+02	-2.64E+02
	vtlin(V)	-5.48E-01	-4.63E-01	-4.62E-01	-4.61E-01	-5.12E-01	-4.43E-01	-4.42E-01	-4.41E-01
	vtsat(V)	-5.13E-01	-4.05E-01	-4.04E-01	-4.03E-01	-4.78E-01	-3.87E-01	-3.86E-01	-3.83E-01
	body_effect(V)	-1.07E+00	-8.01E-01	-8.01E-01	-8.01E-01	-1.06E+00	-7.85E-01	-7.85E-01	-7.85E-01
	ioff(pA/um)	-7.15E+00	-2.04E+02	-2.06E+02	-2.11E+02	-1.46E+01	-2.90E+02	-2.94E+02	-3.03E+02
TT	ft(GHz)	2.2376	11.457	11.317	10.923	NA	NA	NA	NA
	idsat(uA/um)	-7.91E+01	-2.24E+02	-2.24E+02	-2.22E+02	-7.71E+01	-2.42E+02	-2.41E+02	-2.39E+02
	vtlin(V)	-5.80E-01	-5.06E-01	-5.07E-01	-5.08E-01	-5.45E-01	-4.86E-01	-4.86E-01	-4.87E-01
	vtsat(V)	-5.44E-01	-4.49E-01	-4.49E-01	-4.50E-01	-5.09E-01	-4.29E-01	-4.29E-01	-4.29E-01
	body_effect(V)	-1.07E+00	-8.15E-01	-8.15E-01	-8.14E-01	-1.06E+00	-7.99E-01	-7.99E-01	-7.98E-01
	ioff(pA/um)	-3.94E+00	-9.09E+01	-9.02E+01	-8.76E+01	-8.12E+00	-1.32E+02	-1.32E+02	-1.29E+02
FNFP	ft(GHz)	2.1298	10.396	10.269	9.9119	NA	NA	NA	NA
	idsat(uA/um)	-7.41E+01	-2.04E+02	-2.03E+02	-2.02E+02	-7.20E+01	-2.18E+02	-2.17E+02	-2.15E+02
	vtlin(V)	-6.13E-01	-5.50E-01	-5.51E-01	-5.55E-01	-5.77E-01	-5.28E-01	-5.30E-01	-5.34E-01
	vtsat(V)	-5.76E-01	-4.92E-01	-4.93E-01	-4.97E-01	-5.40E-01	-4.71E-01	-4.72E-01	-4.75E-01
	body_effect(V)	-1.07E+00	-8.29E-01	-8.28E-01	-8.27E-01	-1.06E+00	-8.12E-01	-8.12E-01	-8.11E-01
	ioff(pA/um)	-2.17E+00	-4.07E+01	-3.96E+01	-3.67E+01	-4.53E+00	-6.09E+01	-5.94E+01	-5.56E+01
FF	ft(GHz)	2.3994	13.433	13.277	12.833	NA	NA	NA	NA
	idsat(uA/um)	-8.90E+01	-2.64E+02	-2.64E+02	-2.64E+02	-8.76E+01	-2.95E+02	-2.95E+02	-2.93E+02
	vtlin(V)	-5.17E-01	-4.18E-01	-4.16E-01	-4.13E-01	-4.81E-01	-3.98E-01	-3.96E-01	-3.92E-01
	vtsat(V)	-4.83E-01	-3.58E-01	-3.56E-01	-3.51E-01	-4.47E-01	-3.38E-01	-3.36E-01	-3.31E-01
	body_effect(V)	-1.04E+00	-7.35E-01	-7.35E-01	-7.35E-01	-1.03E+00	-7.13E-01	-7.13E-01	-7.14E-01
	ioff(pA/um)	-1.15E+01	-4.69E+02	-4.84E+02	-5.20E+02	-2.36E+01	-6.73E+02	-6.97E+02	-7.56E+02

Table 2-6 RF & Logic Corner Table at 125 °C

Temp=-50C		RF Model				Logic Model			
Device	LF (um)	1	0.34	0.34	0.34	1	0.34	0.34	0.34
	WF (um)	10	15	10	5	10	15	10	5
	NF	16	16	16	16	16	16	16	16
	M	1	1	1	1	1	1	1	1
SS	ft(GHz)	2.6312	11.711	11.567	11.165	NA	NA	NA	NA
	idsat(uA/um)	-9.32E+01	-2.33E+02	-2.32E+02	-2.30E+02	-9.14E+01	-2.50E+02	-2.49E+02	-2.46E+02
	vtlin(V)	-8.26E-01	-7.73E-01	-7.75E-01	-7.81E-01	-7.88E-01	-7.48E-01	-7.50E-01	-7.55E-01
	vtsat(V)	-8.06E-01	-7.38E-01	-7.40E-01	-7.45E-01	-7.68E-01	-7.12E-01	-7.14E-01	-7.18E-01
	body_effect(V)	-1.03E+00	-8.39E-01	-8.38E-01	-8.36E-01	-1.02E+00	-8.23E-01	-8.23E-01	-8.21E-01
	ioff(pA/um)	-2.37E-03	-2.11E-03	-2.37E-03	-5.13E-03	-7.30E-02	-4.84E-02	-7.30E-02	-1.47E-01
SNFP	ft(GHz)	2.9968	15.017	14.85	14.376	NA	NA	NA	NA
	idsat(uA/um)	-1.13E+02	-3.04E+02	-3.04E+02	-3.04E+02	-1.13E+02	-3.45E+02	-3.45E+02	-3.44E+02
	vtlin(V)	-7.35E-01	-6.46E-01	-6.45E-01	-6.43E-01	-6.97E-01	-6.24E-01	-6.23E-01	-6.21E-01
	vtsat(V)	-7.16E-01	-6.06E-01	-6.05E-01	-6.03E-01	-6.78E-01	-5.85E-01	-5.84E-01	-5.81E-01
	body_effect(V)	-1.00E+00	-7.32E-01	-7.31E-01	-7.31E-01	-9.90E-01	-7.17E-01	-7.16E-01	-7.16E-01
	ioff(pA/um)	-2.76E-03	-2.11E-03	-3.55E-03	-5.92E-03	-7.30E-02	-5.16E-02	-7.66E-02	-1.50E-01
TT	ft(GHz)	2.8487	13.793	13.639	13.203	NA	NA	NA	NA
	idsat(uA/um)	-1.06E+02	-2.79E+02	-2.79E+02	-2.78E+02	-1.06E+02	-3.12E+02	-3.11E+02	-3.10E+02
	vtlin(V)	-7.66E-01	-6.87E-01	-6.87E-01	-6.88E-01	-7.28E-01	-6.65E-01	-6.65E-01	-6.65E-01
	vtsat(V)	-7.46E-01	-6.48E-01	-6.48E-01	-6.49E-01	-7.08E-01	-6.26E-01	-6.26E-01	-6.25E-01
	body_effect(V)	-1.00E+00	-7.46E-01	-7.46E-01	-7.45E-01	-9.91E-01	-7.31E-01	-7.31E-01	-7.30E-01
	ioff(pA/um)	-2.76E-03	-2.11E-03	-2.76E-03	-3.55E-03	-7.30E-02	-4.95E-02	-7.38E-02	-1.47E-01
FNFP	ft(GHz)	2.6907	12.486	12.346	11.95	NA	NA	NA	NA
	idsat(uA/um)	-9.90E+01	-2.53E+02	-2.53E+02	-2.51E+02	-9.81E+01	-2.78E+02	-2.78E+02	-2.76E+02
	vtlin(V)	-7.97E-01	-7.29E-01	-7.30E-01	-7.33E-01	-7.59E-01	-7.05E-01	-7.07E-01	-7.10E-01
	vtsat(V)	-7.77E-01	-6.90E-01	-6.91E-01	-6.94E-01	-7.39E-01	-6.66E-01	-6.67E-01	-6.70E-01
	body_effect(V)	-1.00E+00	-7.60E-01	-7.59E-01	-7.58E-01	-9.92E-01	-7.45E-01	-7.44E-01	-7.43E-01
	ioff(pA/um)	-2.76E-03	-2.11E-03	-2.76E-03	-3.55E-03	-7.30E-02	-4.89E-02	-7.30E-02	-1.47E-01
FF	ft(GHz)	3.0258	15.708	15.543	15.07	NA	NA	NA	NA
	idsat(uA/um)	-1.19E+02	-3.21E+02	-3.22E+02	-3.23E+02	-1.20E+02	-3.72E+02	-3.73E+02	-3.73E+02
	vtlin(V)	-7.06E-01	-6.03E-01	-6.01E-01	-5.97E-01	-6.68E-01	-5.81E-01	-5.79E-01	-5.74E-01
	vtsat(V)	-6.88E-01	-5.61E-01	-5.59E-01	-5.53E-01	-6.50E-01	-5.39E-01	-5.37E-01	-5.31E-01
	body_effect(V)	-9.70E-01	-6.64E-01	-6.64E-01	-6.65E-01	-9.60E-01	-6.44E-01	-6.44E-01	-6.45E-01
	ioff(pA/um)	-1.97E-03	-6.84E-03	-8.29E-03	-9.87E-03	-7.30E-02	-6.05E-02	-8.57E-02	-1.62E-01

Table 2-7 RF & Logic Corner Table at -50 °C

2.2.2 Summary Table

In the following fitting plots & summary table, the parasitic resistances due to metal interconnection from pad to DUT (Device Under Test) are taken into account. Since RF DUT generally use larger total gate width and has larger current, these parasitic resistances will cause voltage drops. Therefore the extracted I_{dsat} is smaller than the real I_{dsat} for the DUT without the parasitic resistance.

Device						Vtlin(V)		Vtsat(V)		Idsat(uA/um)		Ioff(pA/um)		ft(GHz)	
Name	LF(um)	WF(um)	NF	M	con	Measure	Model	Measure	Model	Measure	Model	Measure	Model	Measure	Model
PFET801	0.3	2.5	16	5	1	-0.584	-0.584	-0.519	-0.529	-286.61	-284.03	-167.5	-4.372	12.978	13.532
PFET802	0.3	5	16	4	1	-0.586	-0.58	-0.521	-0.526	-286.61	-283.98	-112.14	-2.94	14.054	14.113
PFET803	0.3	10	16	2	1	-0.585	-0.578	-0.52	-0.525	-282.81	-283.83	-107.78	-2.223	14.541	14.766
PFET804	0.3	15	16	1	1	-0.583	-0.578	-0.516	-0.525	-280.65	-283.03	-145.31	-1.982	14.672	14.914
PFET805	0.3	10	4	8	1	-0.582	-0.578	-0.516	-0.526	-299.84	-294.65	-104.44	-2.336	14.976	14.841
PFET807	0.3	10	32	1	1	-0.595	-0.578	-0.515	-0.525	-278.66	-281.46	-112.55	-2.204	14.301	14.672
PFET808	0.3	5	2	4	1	-0.598	-0.58	-0.518	-0.528	-317.75	-309.05	-860.6	-3.524	14.738	14.726
PFET810	0.3	5	8	2	1	-0.6	-0.58	-0.52	-0.526	-296.18	-288.72	-440.29	-3.024	14.571	14.457
PFET811	0.3	5	11	1	1	-0.608	-0.58	-0.525	-0.526	-295.35	-286.86	-643	-2.978	14.752	14.426
PFET812	0.3	5	32	1	1	-0.603	-0.58	-0.522	-0.526	-286.29	-283.05	-214.58	-2.898	14.311	14.358
PFET813	0.34	10	16	1	1	-0.625	-0.607	-0.558	-0.562	-254.86	-248.86	-209.98	-1.194	12.704	12.409
PFET814	0.5	10	16	2	1	-0.663	-0.656	-0.613	-0.623	-173.78	-170.55	-108.22	-0.822	7.257	7.018
PFET815	0.7	10	16	2	1	-0.675	-0.673	-0.633	-0.646	-127.07	-125.69	-107.25	-0.796	4.396	4.253
PFET816	1	10	16	3	1	-0.681	-0.683	-0.642	-0.658	-90.27	-91.283	-72.03	-0.789	2.501	2.515
PFET840	0.36	5	16	4	1	-0.633	-0.619	-0.571	-0.575	-238.98	-234.2	-107.18	-1.815	11.215	10.972
PFET846	0.36	5	2	4	1	-0.631	-0.618	-0.576	-0.576	-263.34	-250.96	-850.22	-2.449	11.757	11.222
PFET847	0.36	5	4	2	1	-0.64	-0.618	-0.576	-0.576	-254.88	-241.79	-848.96	-2.086	11.944	11.14
PFET848	0.36	5	8	2	1	-0.638	-0.618	-0.574	-0.575	-246.58	-237.17	-438.21	-1.906	11.667	11.104
PFET850	0.36	5	32	1	1	-0.633	-0.619	-0.57	-0.575	-238.53	-233.35	-213.78	-1.769	11.239	11.073

Table 2-8 Summary Table for RF PMOS

Measure: Measurement result

Model: Simulation result

2.2.3 Simulation results for WF, LF trends

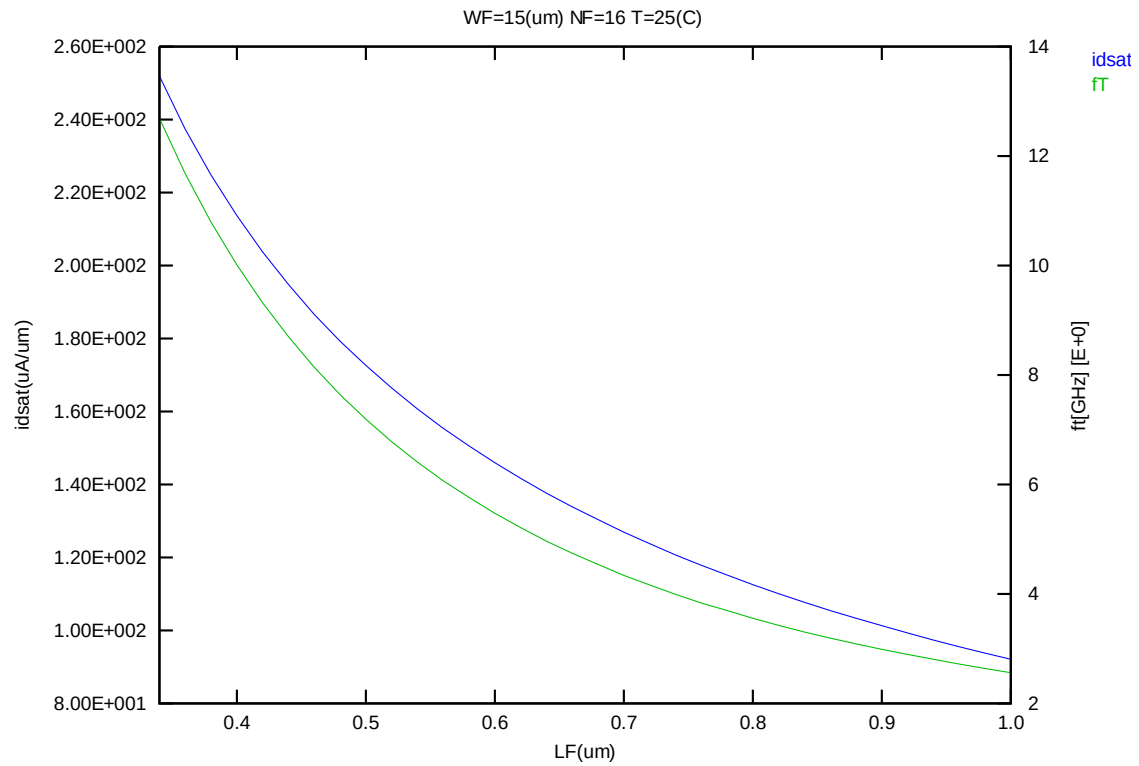


Figure 2-14 fT and Idsat vs. channel length

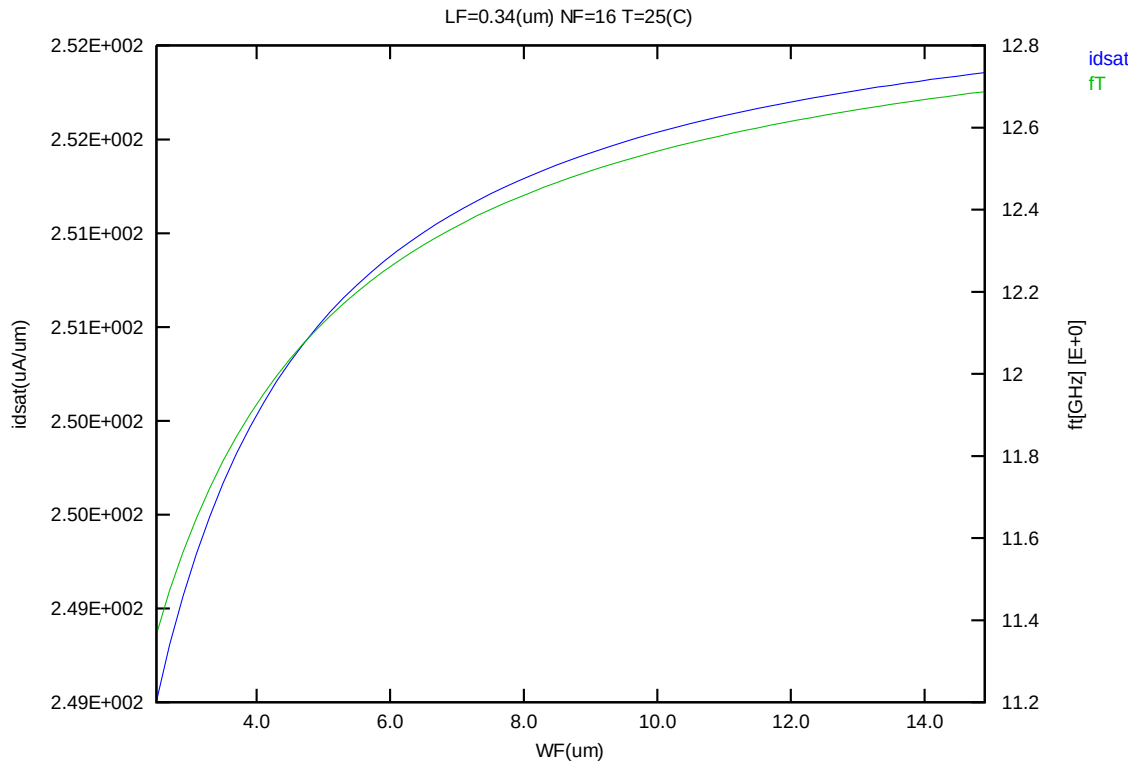


Figure 2-15 f_T and I_{dsat} vs. channel width

2.2.4 DC & RF Fitting curves for individual devices

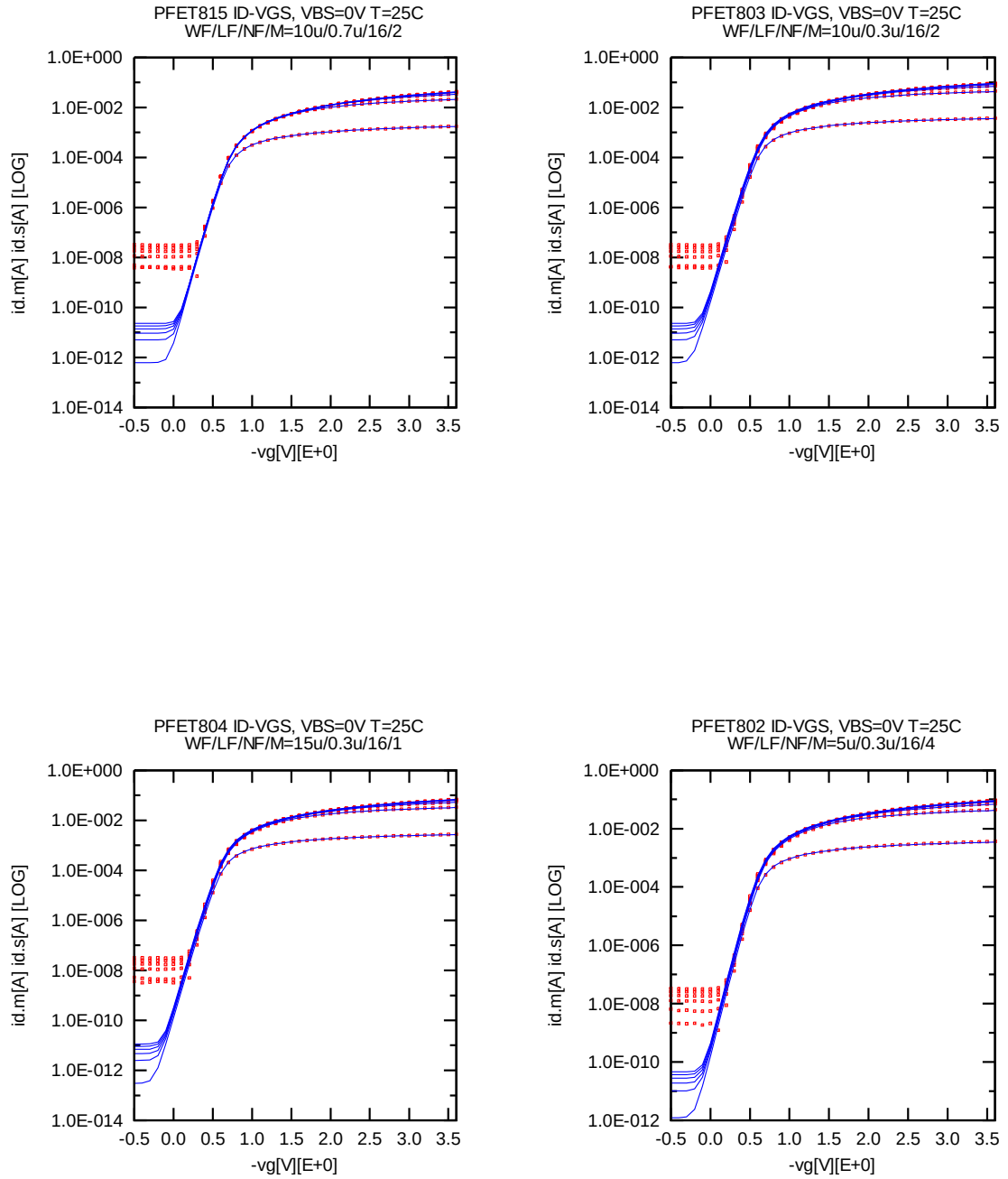


Figure 2-16 Id-Vg (25°C)

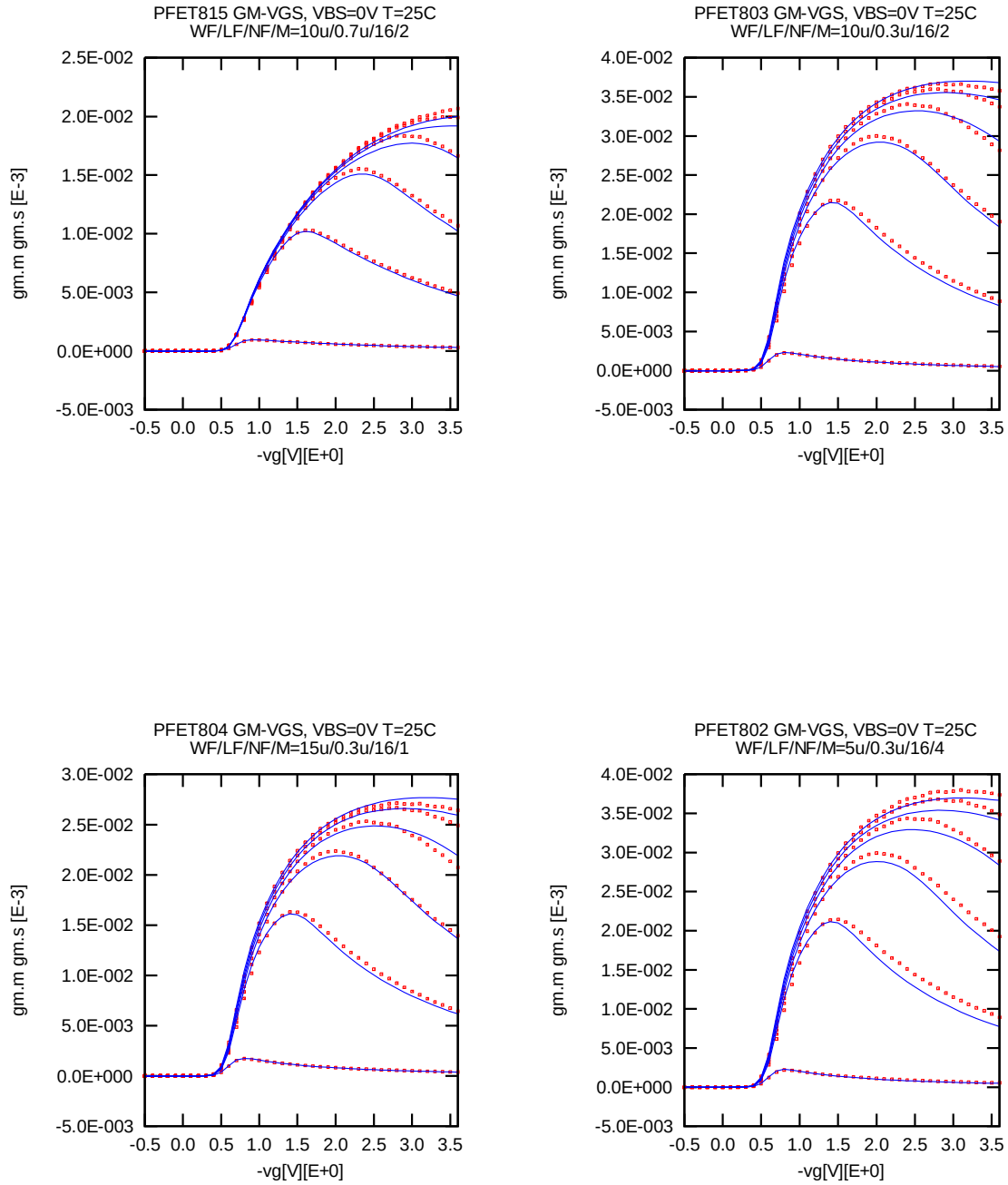


Figure 2-17 Gm-Vg (25°C)

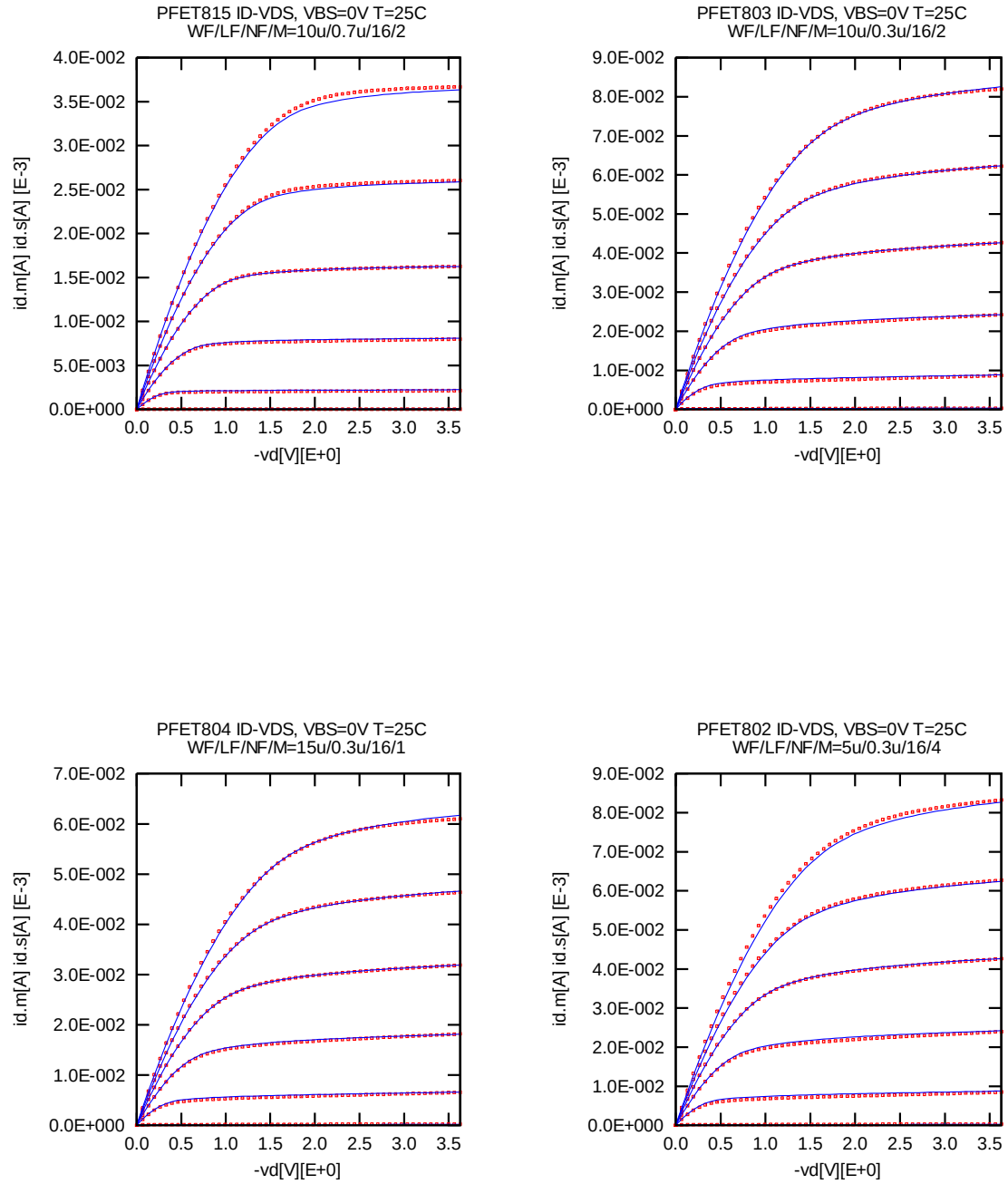


Figure 2-18 Id-Vd (25°C)

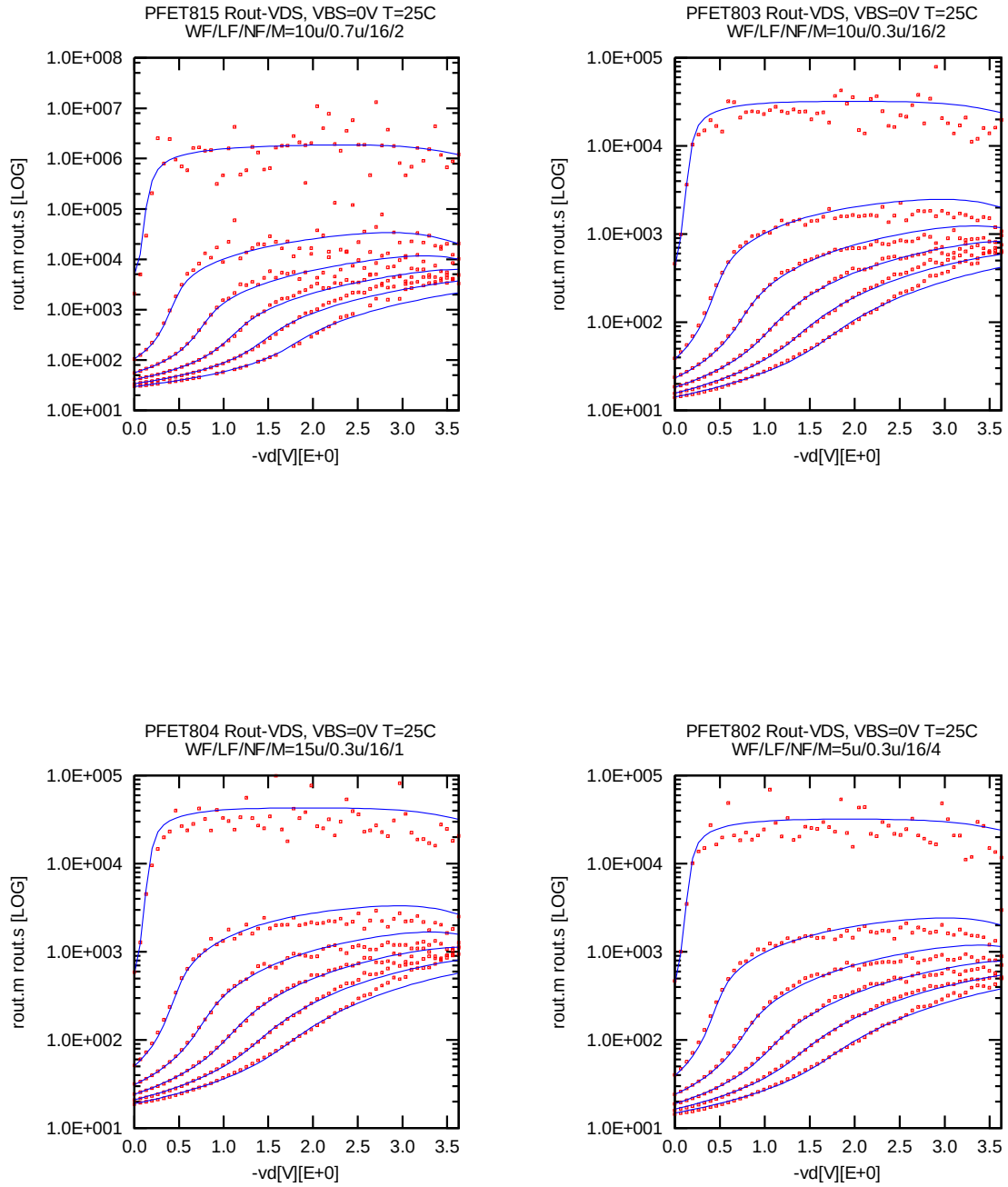


Figure 2-19 Rout-Vd (25°C)

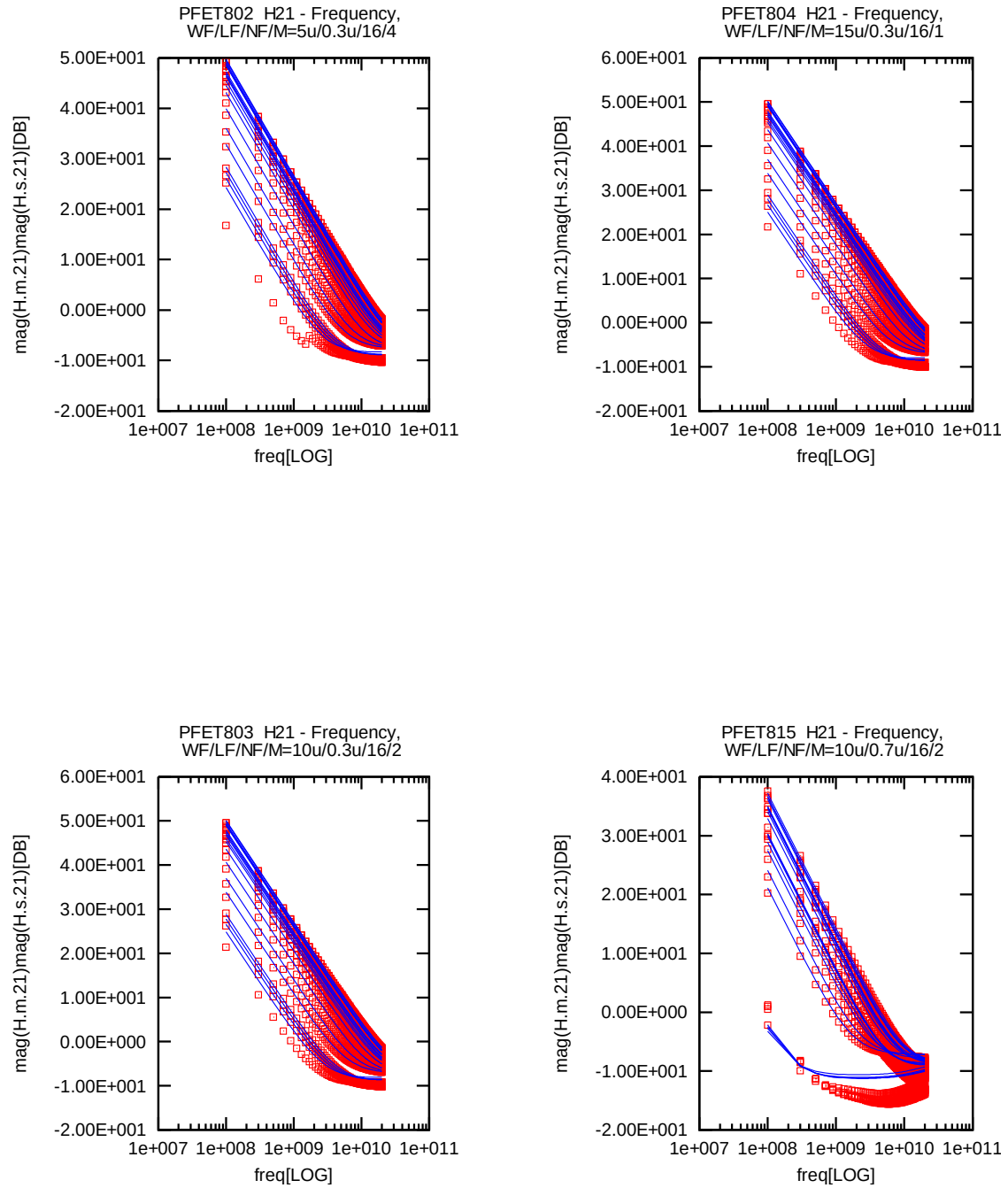


Figure 2-20 Mag(H21)-Freq (25°C)

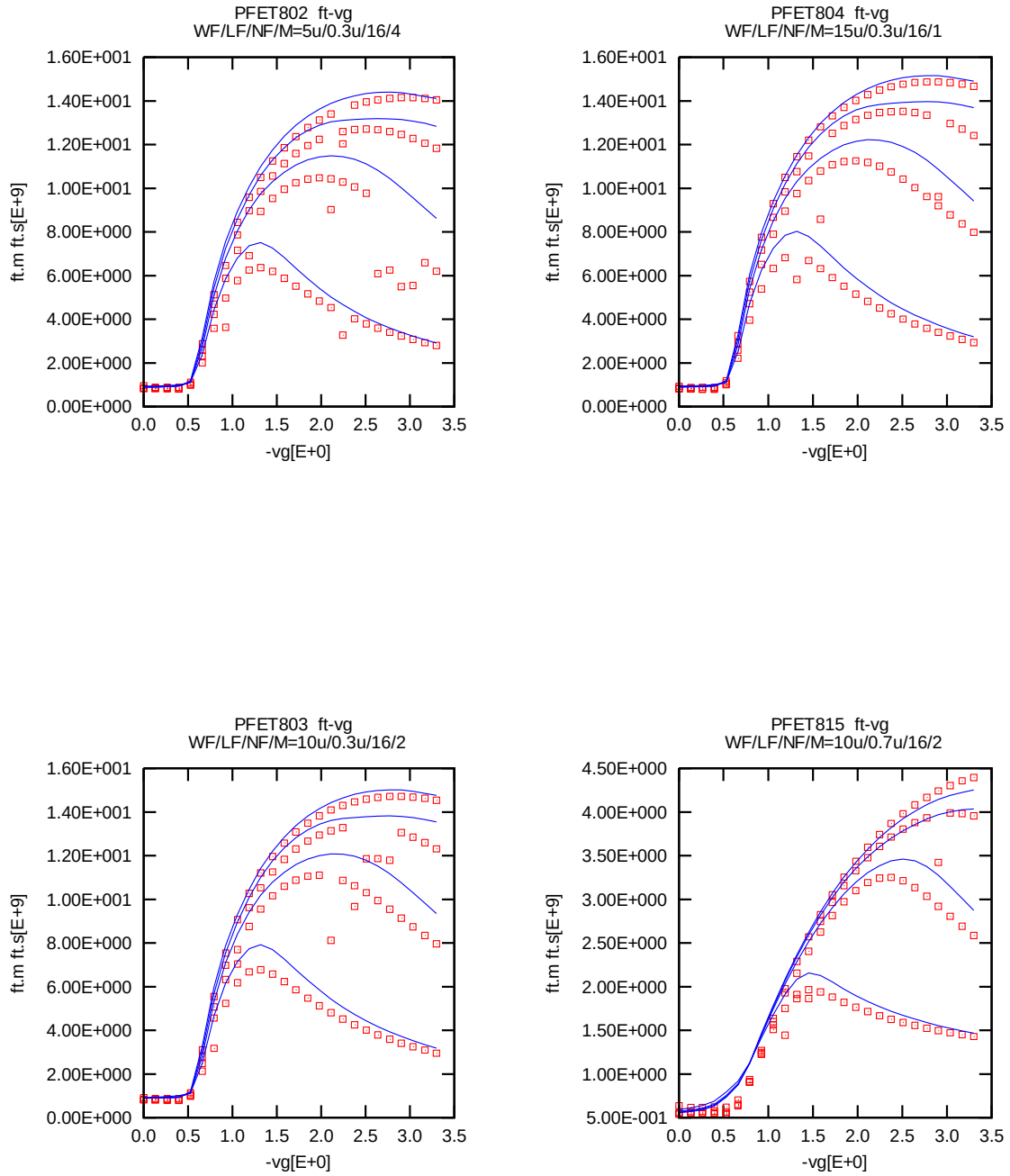


Figure 2-21 f_t - V_g (25°C)

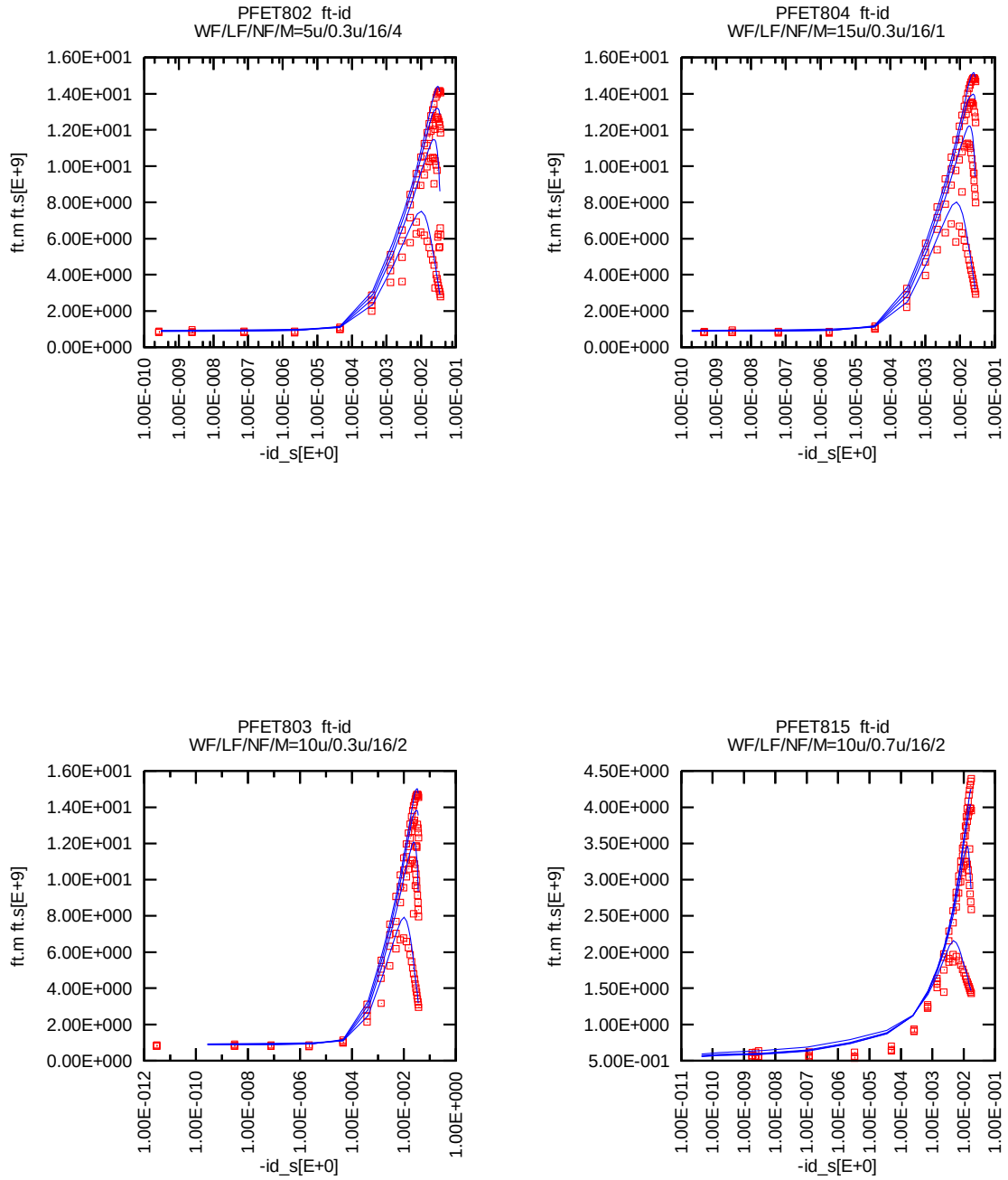


Figure 2-22 f_T -Id (25°C)

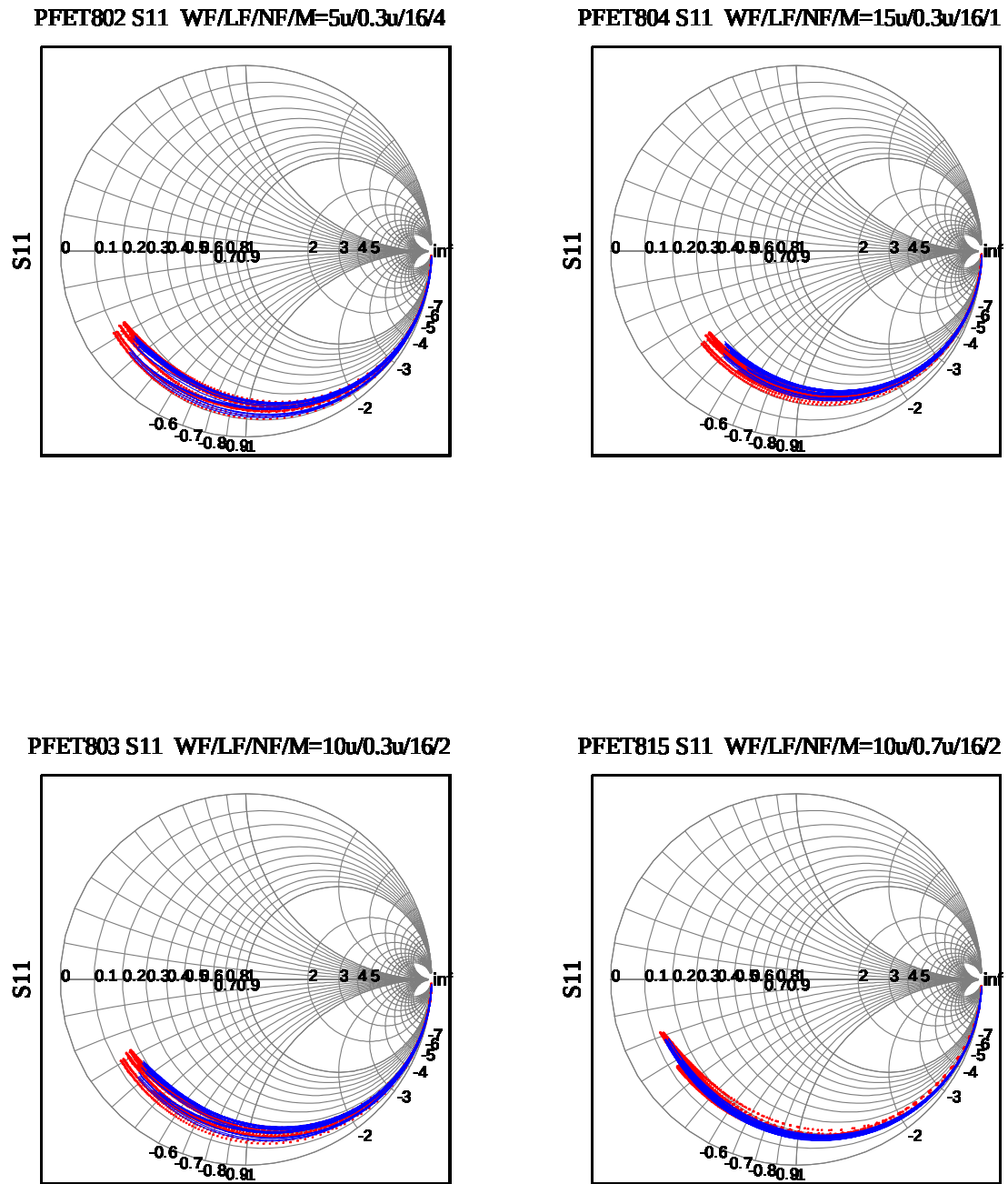


Figure 2-23 S11 (25°C)

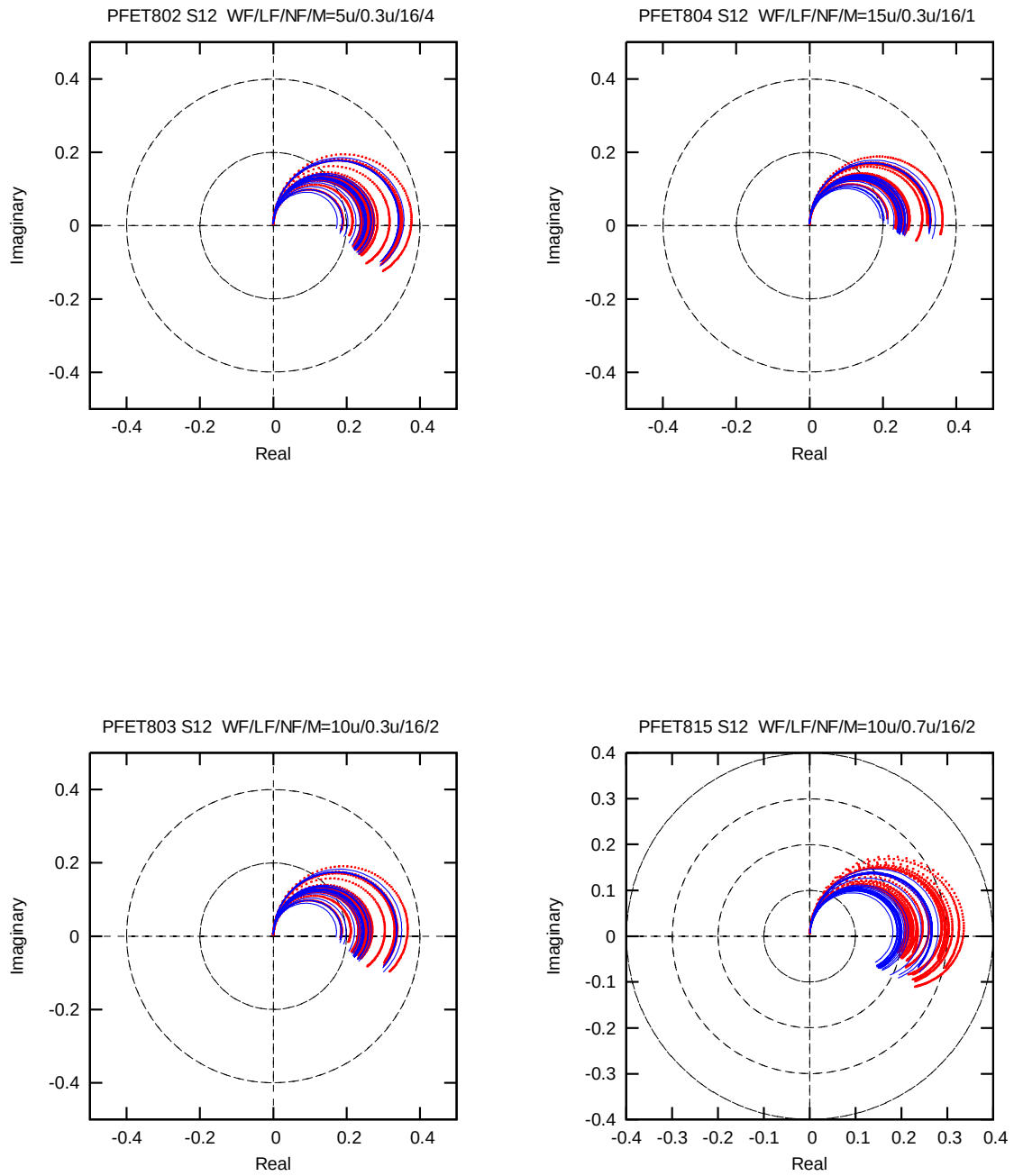


Figure 2-24 S12 (25°C)

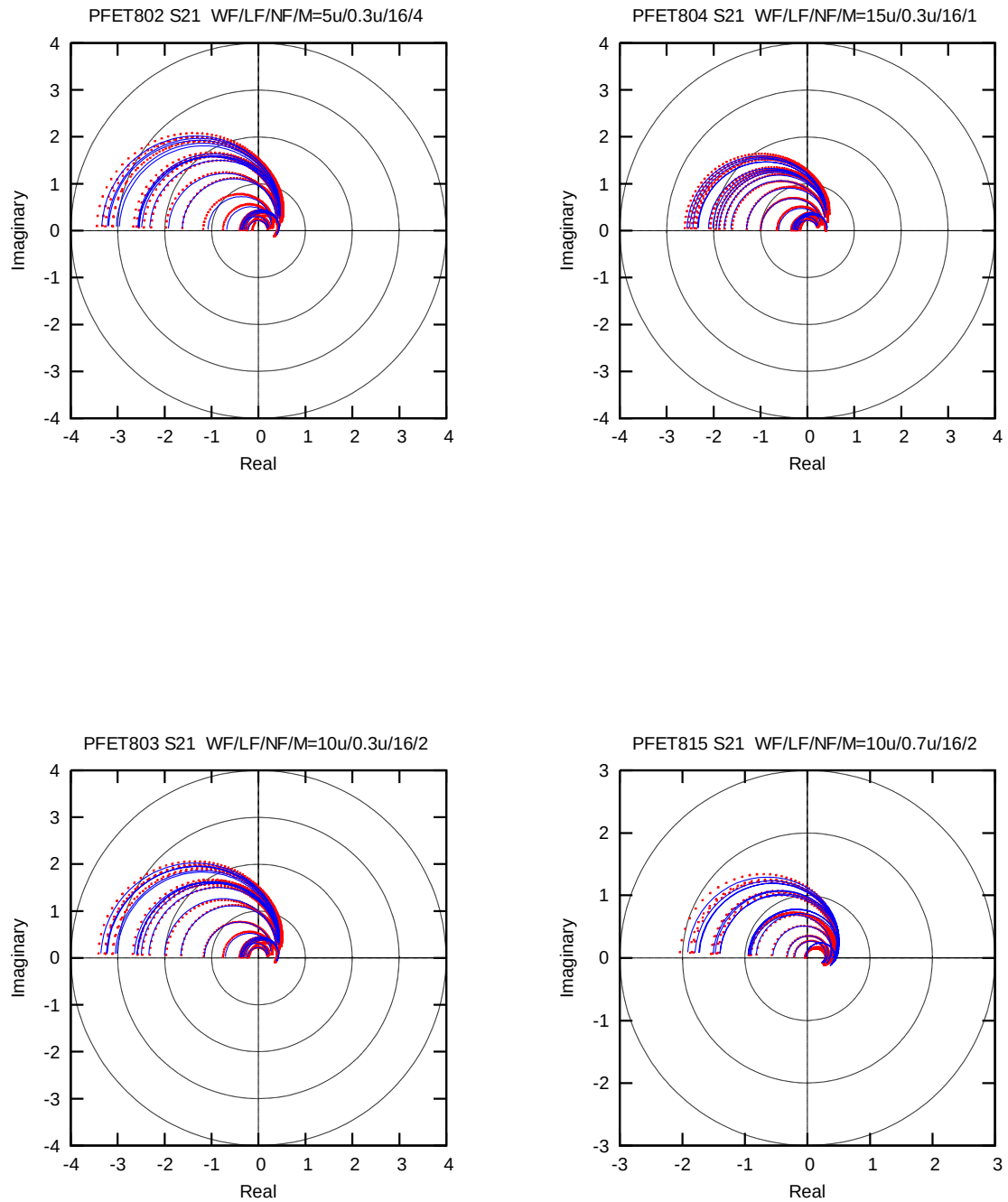


Figure 2-25 S21 (25°C)

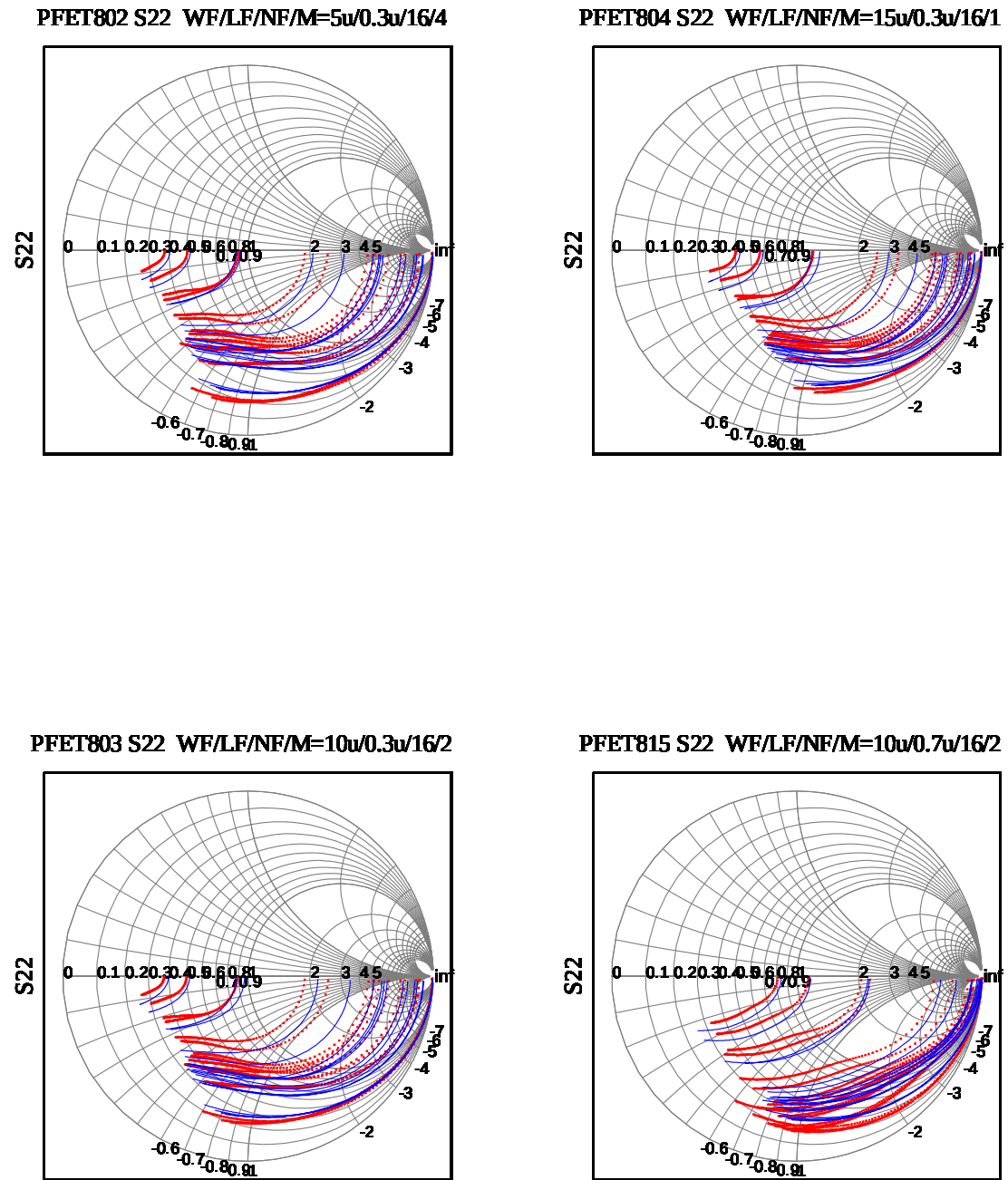


Figure 2-26 S22 (25°C)

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

Item	Warning Message	Explanation
1.	None	None

3.2 SPECTRE Warning Message

Table 3-2 SPECTRE warning message

Item	Warning Message	Explanation
1.	None	None

3.3 ELDO Warning Message

Table 3-3 ELDO warning message

Item	Warning Message	Explanation
1.	None	None